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May 06, 2025

Division of Food Ingredients
Office of Pre-Market Additive Safety
Office of Food Chemical Safety, Dietary Supplements, and Innovation
U.S. Food and Drug Administration
5001 Campus Dr.
College Park, MD 20740

Attention: Dr. Susan Carlson
Re: GRAS Notification – Autoclaved Parabacteroides goldsteinii RV-01

Dear Dr. Carlson:

GRAS Associates, LLC, acting as the Agent for ReviveBio, Co. (China), is submitting for FDA review Form 3667 and a GRAS Notification for Autoclaved Parabacteroides goldsteinii RV-01. Along with ReviveBio, Co.'s determination of safety, an Expert Panel of qualified persons was assembled to assess the composite safety information of the subject substance under the intended use as an ingredient in sports and energy drinks, ready-to-eat cereals (low sugar), yogurt, fermented milk products, ice cream, cereal bars, nutrition bars, nut spreads, and candy containing chocolate at levels up to 1×10^{10} TFU/serving, or 185 mg/serving. The intended use does not include use in infant formula or USDA-regulated food. The mean and 90th percentile intakes of autoclaved P. goldsteinii RV-01 in the general population are 223.8 mg/day (1.0×10^{10} TFU/day) and 475.9 mg/day (2.2×10^{10} TFU/day), respectively. The notification contains the specific information that addresses the safe human food uses for the subject notified substance as discussed in the GRAS guidance document.

If additional information or clarification is needed as you and your colleagues proceed with the review, please feel free to contact me via telephone or email. I also authorize Amy Mozingo (amozingo@gras-associates.com), VP US Nutra Regulatory Sciences, GRAS Associates LLC to lead communications related to this submission.

We look forward to your feedback.

Sincerely,


William J. Rowe

Agent for ReviveBio, Co.

President & CEO
GRAS Associates, LLC
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Independent GRAS Conclusion

of

Autoclaved *Parabacteroides goldsteinii* RV-01

Food Usage Conditions for General Recognition of Safety

on behalf of

ReviveBio, Co.

**14F-5, No. 3, Park St., NangGang District, Taipei City, 11503,
Taiwan (R.O.C.)**

4/22/25

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FOREWORD

ReviveBio, Co. based our Generally Recognized as Safe (GRAS) assessment of Autoclaved *Parabacteroides goldsteinii* RV-01 on the composite safety information, i.e., scientific procedures. The safety/toxicity of autoclaved *Parabacteroides goldsteinii* RV-01, history of use of *Parabacteroides goldsteinii* RV-01, and compositional details, specifications, and method of preparation of the subject ingredient were reviewed. In addition, a search of scientific and regulatory literature was conducted through February 10, 2025, with particular attention paid to adverse reports, as well as those that pertain to the overall conclusions of safety. Those references deemed pertinent to this review are listed in Part 7. The composite safety/toxicity studies, in concert with dietary exposure information, ultimately provide the specific scientific foundation for the GRAS conclusion.

At ReviveBio, Co.'s request, GRAS Associates, LLC ("GA") convened an Expert Panel to complete an independent safety evaluation of ReviveBio, Co.'s autoclaved *Parabacteroides goldsteinii* RV-01 ingredient. The purpose of the evaluation is to ascertain whether ReviveBio, Co.'s Autoclaved *Parabacteroides goldsteinii* RV-01 is generally recognized as safe, i.e., GRAS, under the intended conditions of use.

PART 1. SIGNED STATEMENTS AND CERTIFICATION

ReviveBio, Co. has concluded that our Autoclaved *Parabacteroides goldsteinii* RV-01, referred to as “autoclaved *P. goldsteinii* RV-01” and which meets the specifications described below, is GRAS in accordance with Section 201(s) of the Federal Food, Drug, and Cosmetic Act (FD&C Act). This determination was made in concert with an appropriately convened panel of experts who are qualified by scientific training and experience. The GRAS determination is based on scientific procedures as described in the following sections. The evaluation accurately reflects the intended conditions of food use for the designated autoclaved *P. goldsteinii* RV-01 preparation.

This signed statement and certification have been prepared in accordance with the requirements of 21 CFR 170.225.

This certification is signed by a responsible official of GRAS Associates, LLC acting as agent for ReviveBio, Co.

This Part 1 of the GRAS dossier does not include any confidential information.

This Independent GRAS Assessment was conducted in accordance with Subpart E of 21 CFR Part 170.

Names and addresses of organizations;

Sponsoring Party:

ReviveBio, Co.

14F-5, No. 3, Park St., NangGang District, Taipei City, 11503, Taiwan (R.O.C.)

As the Responsible Party, ReviveBio, Co. accepts responsibility for the GRAS conclusion that has been made for Autoclaved *Parabacteroides goldsteinii* RV-01 ingredient, as described in the subject safety evaluation.

Agent:

GRAS Associates, LLC

11810 Grand Park Avenue

Suite 500

North Bethesda, MD 20852

The name of the ingredient is: Autoclaved *P. goldsteinii* RV-01.

The ingredient is intended to be used in sports and energy drinks, ready-to-eat cereals (low sugar), yogurt, fermented milk products, ice cream, cereal bars, nutrition bars, nut spreads, and candy containing chocolate at levels up to 1×10^{10} total fluorescent units TFU /serving. The ingredient is not intended to be used in infant formula or USDA-regulated foods.

The statutory basis for our conclusion of GRAS status is through scientific procedures in accordance with § 170.30(a) and (b).

It is our view that the ingredient is not subject to the premarket approval requirements of the FD&C Act based on our conclusion that the substance is GRAS under the conditions of its intended use.

If the U.S. Food and Drug Administration (FDA) were to ask to see the data and information that are the basis for our conclusion of GRAS status, we agree to:

- (i) make the data and information available to FDA; and
- (ii) agree to both of the following procedures for making the data and information available to FDA:
 - (A) Upon FDA's request, we will allow FDA to review and copy the data and information during customary business hours at our address specified where these data and information will be available; and
 - (B) Upon request by FDA, we will provide FDA with a complete copy of the data and information either in an electronic format that is accessible for their evaluation or on paper.

We certify that, to the best of our knowledge, this GRAS Assessment is a complete, representative, and balanced review that includes unfavorable information, as well as favorable information, known to us and pertinent to the evaluation of the safety and GRAS status of the use of the substance.

ReviveBio, Co. does not intend to add autoclaved *P. goldsteinii* RV-01 to any products that fall under FSIS/USDA jurisdiction. Therefore, 21 CFR 170.270 does not apply.

Signature



Agent for ReviveBio, Co.

William J. Rowe
President
GRAS Associates, LLC
11810 Grand Park Ave
Suite 500
North Bethesda, MD 20852

Date: April 22, 2025

PART 2. IDENTITY, METHOD OF MANUFACTURE, SPECIFICATIONS, AND PHYSICAL OR TECHNICAL EFFECT

A. Chemical Identity of Ingredient

“*Parabacteroides goldsteinii* RV-01” is the common or usual name of the live microbial from which the autoclaved ingredient is derived. The compositional features of *P. goldsteinii* RV-01 are described in more detail in this section. “Autoclaved *P. goldsteinii* RV-01” is the term used by ReviveBio, Co. in referring to the ingredient. The ingredient is also intended to be marketed as “REVIVEPRO™”.

Table 1. Taxonomic Classification of *P. goldsteinii* RV-01

Kingdom	Bacteria
Phylum	<i>Bacteroidota</i>
Class	<i>Bacteroidia</i>
Order	<i>Bacteroidales</i>
Family	<i>Tannerellaceae</i>
Genus	<i>Parabacteroides</i>
Species	<i>goldsteinii</i>
Strain	RV-01

Reference: [Taxonomy browser \(*Parabacteroides goldsteinii*\) \(nih.gov\)](#)

B. Phenotypic Identity

Parabacteroides goldsteinii RV-01 is a strictly anaerobic, gram-negative, non-spore forming, rod-shaped bacterium belonging to the Bacteroidota phylum. Colonies on agar plates are 1-2 mm in diameter, grey to off-white–grey, circular, slightly convex and smooth. *Parabacteroides goldsteinii* are saccharolytic bacteria, and major end-products are typically acetic and succinic acids; minor amounts of isovaleric acid, propionic acid and formic acid are also produced. *Parabacteroides goldsteinii* can also grow on medium containing 20% bile (Sakamoto and Benno, 2006). Further, *P. goldsteinii* RV-01 is reported to be non-pathogenic (Lin et al., 2024).

C. Genotypic Identity

The identification and grouping of *P. goldsteinii* RV-01 was achieved by a 16S rRNA sequence analysis. Due to the full length 16S rRNA gene sequence showing 99.71% identity to the *P. goldsteinii* JCM13446 in the NCBI Basic Local Alignment Search Tool (BLAST) analysis¹, the isolated bacterial strain was identified as *P. goldsteinii* RV-01. The whole genome sequence and annotation of *P. goldsteinii* RV-01 has been deposited at NCBI under the BioProject number PRJNA1153404. Further, the genome of *P. goldsteinii* RV-01 has been deposited in the Japanese National Institute of

¹ Available online at: https://blast.ncbi.nlm.nih.gov/Blast.cgi?PROGRAM=blastn&PAGE_TYPE=BlastSearch&LINK_LOC=blasthome

Technology and Evaluation (NITE) under accession number BP-04231. The circular genome of *P. goldsteinii* RV-01 with annotation has also been published by Lin et al. (2024). Briefly, the GC content is 43% and there is no plasmid DNA. Annotation of the sequences revealed the genome consists of 5,241 coding DNA sequences, 79 tRNA genes, and 18 rRNA genes.

In addition, a phylogenetic tree and average nucleotide identity (ANI) analysis were performed, demonstrating the relationship between *P. goldsteinii* RV-01 and other strains of the same genus and species. These analyses are provided in Figure 1 and Table 2. The ANI analysis was completed using the OrthoANLu algorithm described by (Yoon et al., 2017). Typically, an ANI value > 95% indicates that organisms belong to the same species (Jain et al., 2018; Rodriguez-R et al., 2024).

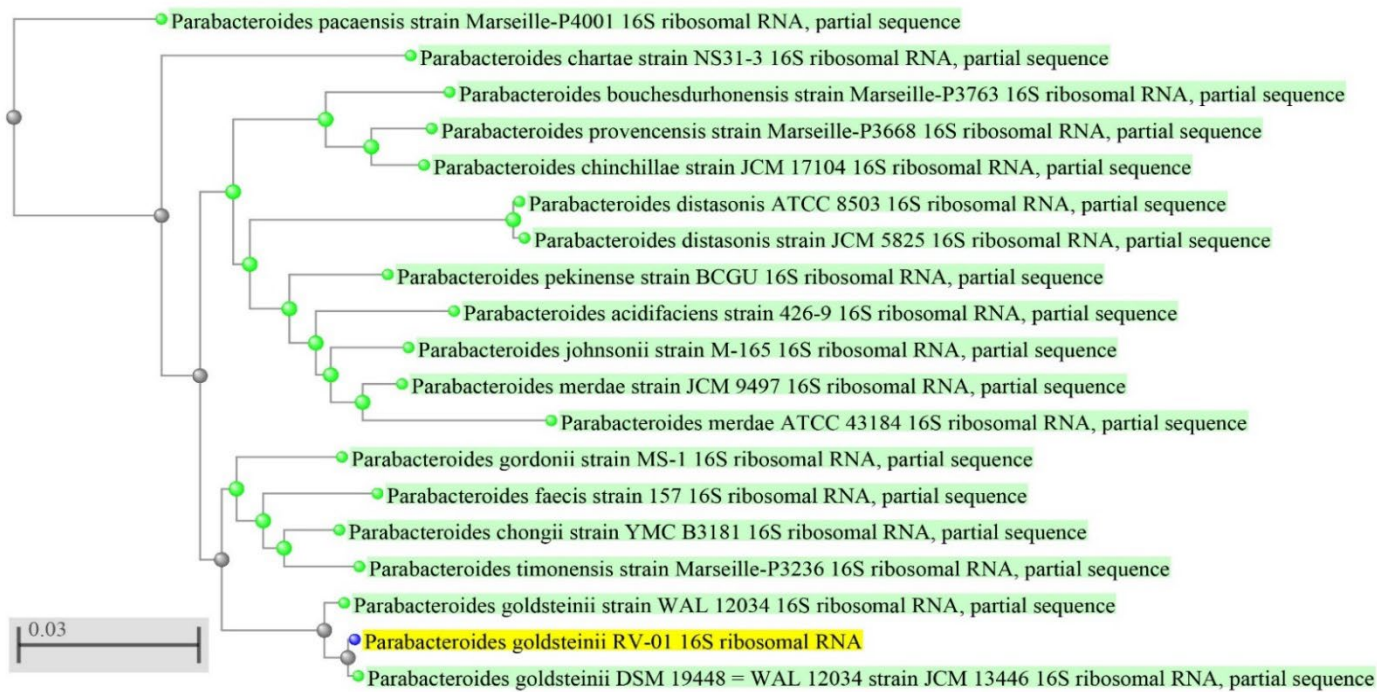


Figure 1. Phylogenetic Tree of *P. goldsteinii* RV-01 and Related Strains

Table 2. ANI Values of *P. goldsteinii* RV-01 and Other *P. goldsteinii* Strains and *Parabacteroides* Species

Strain	ANI Value (%)					
	<i>P. goldsteinii</i> MTS01	<i>P. goldsteinii</i> DSM19448	<i>P. distasonis</i> ATCC8503	<i>P. merdae</i> ATCC43184	<i>P. gordonii</i> DSM23371	<i>P. faecis</i> DSM102983
<i>P. goldsteinii</i> RV-01	98.17	98.44	73.83	76.77	82.57	83.96

ANI – Average nucleotide identity

D. Manufacturing Processes

ReviveBio, Co.'s autoclaved *P. goldsteinii* RV-01 is produced in accordance with current Good Manufacturing Practices (cGMP) according to the Taiwan Ministry of Health and Welfare. Further, the ingredient is manufactured in accordance with ISO 22000:2018. Documentation supporting the supplier's Taiwan cGMP and ISO 22000:2018 compliance is found in Appendix 1, Appendix 2, and Appendix 3. Production of autoclaved *P. goldsteinii* RV-01 is also conducted according to Hazard Analysis and Critical Control Point (HACCP) principles (Appendix 4). All raw materials, processing aids, and food contact substances used to manufacture autoclaved *P. goldsteinii* RV-01 are food-grade ingredients permitted by U.S. regulations or have previously been determined to be GRAS for their respective uses, as attested in Appendix 5.

To avoid genetic drift, a 16S rRNA sequencing-validated working cell bank (WCB) is maintained at the production facility (at -80°C in 10% glycerol) to store inoculum for culture initiation. Briefly, seed culture is initiated using an aliquot from the WCB, and the seed culture is then used to inoculate the production culture in the fermentation vessel where the product organism (*P. goldsteinii* RV-01) is grown. Cells are separated from the growth medium (food grade soy peptones, sodium bicarbonate, sodium chloride, sodium phosphate, sucrose², and cysteine hydrochloride prepared in purified water) by centrifugation, autoclaved at 121°C for 15 minutes, and stored at 4°C. The composition of each batch of culture media is consistent between batches and within batches from seed culture to fermentation culture. A flow chart of the production process is presented below in Figure 2, with further details of each step of the production process outlined below.

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² The sucrose is sourced from sugar cane and/or sugar beet.

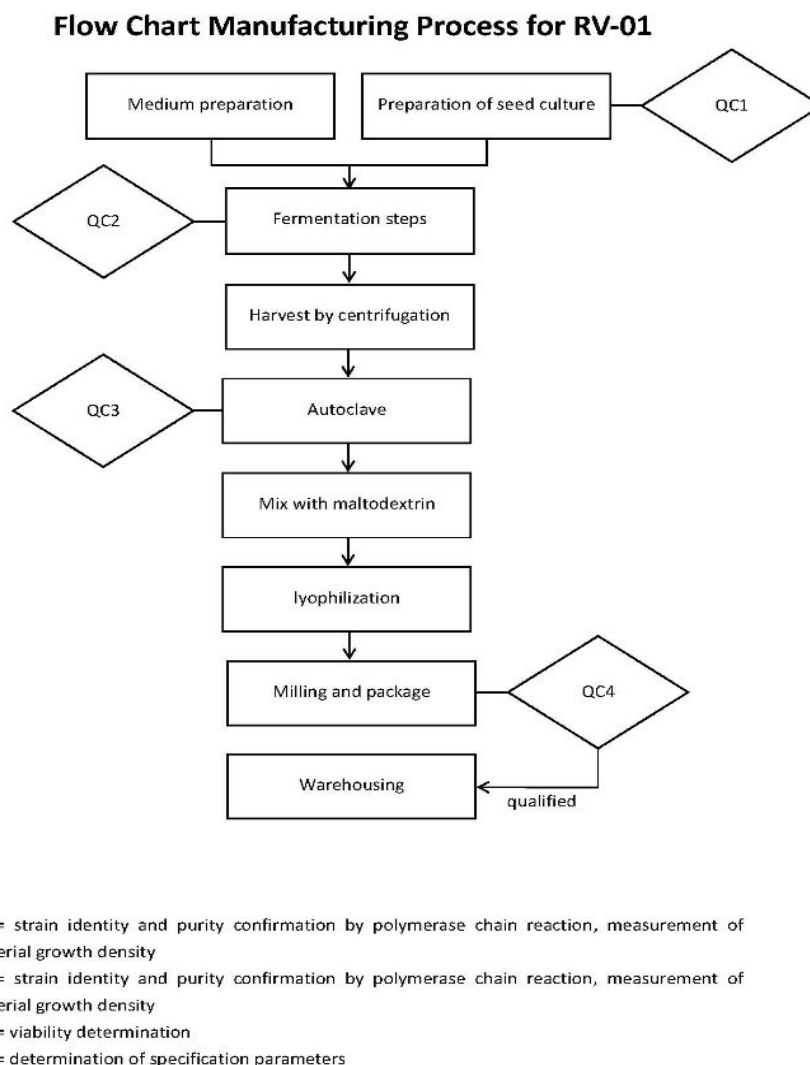


Figure 2. Flow Chart of Manufacturing Process

1. Preparation of Seed Culture

An aliquot of the WCB (1 mL in 10% glycerol) is thawed at 37°C and used to inoculate freshly prepared culture medium (50 – 200 mL). This culture is incubated at 37°C for 24 hours in an anaerobic environment to generate the seed culture. Pre-culture inoculum from the seed culture is added at a ratio of between 1:50 and 1:200, and cells are propagated between 36 and 37°C until end log-phase, as determined by culture optical density. In general, a lower seed culture optical density requires a higher seeding ratio to maintain cell density.

2. Initiation of Fermentation Culture

The production fermenter is prepared with fresh media and inoculated with seed culture at a ratio of 1:50 and 1:200. The fermentation time is dictated by fermentation conditions and the ratio of seed culture to culture medium. The culture undergoes fermentation at 36°C to 37°C until harvesting cell density is reached.

3. Cell Product Harvest

Twenty-four hours after fermentation, cells are harvested by pellet centrifugation for subsequent lyophilization.

4. Autoclaving and Lyophilization

The *P. goldsteinii* RV-01 cells are autoclaved for 15 minutes at 121°C prior to being mixed with maltodextrin (sourced from non-genetically modified corn) at a ratio of 1:6 w/w to achieve optimized cell density and conditions. The resultant product, now referred to as the ingredient, is then freeze-dried in a lyophilizer. This process results in a cake that is then ground to powder for storage. This lyophilized powder is used to prepare the final product. The autoclave and lyophilization processes occur in an aerobic environment.

5. Quality Control

Throughout the production process, line samples³ are taken for quality control analysis to ensure process controls in place are met to guarantee optimized fermentation process reproducibility and efficiency. Quality control (QC) steps are indicated in Figure 3, and involve calibration of all equipment, and monitoring of temperature, pH, and optical density, as well as centrifugation speed. Only if QC requirements are met in each phase is the production process allowed to move forward. This includes the autoclaved *P. goldsteinii* RV-01 ingredient meeting all specifications, including the limit for live *P. goldsteinii* RV-01 cells. The autoclaved *P. goldsteinii* RV-01 ingredient is then packaged in food grade polyethylene packaging.

Ingredients that are major allergens under The Food Allergen Labeling and Consumer Protection Act of 2004 (FALCPA) and the Food Allergy Safety, Treatment, Education and Research Act of 2021 (FASTER) are required to be disclosed and must be declared on product labeling. These include ingredients derived from milk, egg, fish, Crustacean shellfish, tree nuts, wheat, peanuts, soybeans, and sesame. ReviveBio, Co. uses a raw material in their culture media (soy peptone) which is derived

³ Seed culture samples are taken for strain identity and purity confirmation by polymerase chain reaction. Fermentation culture samples are taken for strain identity and purity confirmation by polymerase chain reaction as well as measurement of bacterial growth density. A sample is taken immediately following autoclaving for viability determination. Lastly, a sample of lyophilized powder (containing maltodextrin) is taken for determination of the final specifications.

from soy. Since an ingredient derived from a major allergen (soy) is used, the appropriate allergen declarations will be disclosed for the ingredient and finished products containing the ingredient.

E. Ingredient Specifications and Analysis

While plate counts resulting in colony forming units (CFU) have long been the gold standard for microbial analysis and are very useful for enumerating pathogenic or other undesired microbes, other methods are required to accurately assess the number of dead bacterial cells. Flow cytometry using specific functional dyes has been developed as the method of choice and is used here for the enumeration of *P. goldsteinii* RV-01 (ISO, 2015; Foglia et al., 2020). This method, which is based on membrane integrity (Nebe-Von Caron and Badley, 1995), can be used to monitor the number of dead and intact cells represented as total fluorescent units (TFU). In the specific use described here, total cell count of autoclaved *P. goldsteinii* RV-01 is defined as TFU which represents the sum of dead cells.

The specifications for the autoclaved *P. goldsteinii* RV-01 ingredient and supporting methods of analysis are provided in Table 3.

Table 3. Specifications for Autoclaved *P. goldsteinii* RV-01 and Supporting Methods

Parameter	Autoclaved <i>P. goldsteinii</i> RV-01 Specifications	
	Specification	Method
Appearance	Off-white to beige homogenous powder	Visual
Color	Off-white to beige	Visual
Odor	Characteristic	Olfaction
Moisture (%)	< 5.0	CNS5033
Water Activity	< 2.0	CNS5225
Identification (%)	> 98% <i>P. goldsteinii</i> RV-01	16S rRNA sequencing (Sanger Method)
<i>P. goldsteinii</i> RV-01 Count (TFU/g)	> 1.2×10^{10}	ISO 19344-IDF 232
Viable <i>P. goldsteinii</i> RV-01 Count (CFU/g)*	ND	Internal
Heavy Metals		
Arsenic (ppm)	< 0.1	MOHWM0014.03
Lead (ppm)	< 0.1	MOHWM0014.03
Cadmium (ppm)	< 0.1	MOHWM0014.03
Mercury (ppm)**	< 0.1	MOHWM0014.03
Microorganisms		
Aerobic Plate Count (CFU/g)*	< 500	MOHWM0014.02
Yeast and Molds (CFU/g)*	< 100	MOHWM0008.01
<i>Salmonella</i> (per 25 g)	Absent	MOHWM0025.01
<i>Listeria monocytogenes</i> (CFU/g)*	ND	MOHWM0029.00
<i>Staphylococcus aureus</i> (CFU/g)*	ND	MOHWM0002.02

Parameter	Autoclaved <i>P. goldsteinii</i> RV-01 Specifications	
	Specification	Method
<i>Escherichia coli</i> (MPN/g)***	ND	MOHWM0023.02
<i>Enterobacteriaceae</i> (CFU/g)*	ND	MOHWM0028.00
<i>Bacillus cereus</i> (CFU/g)****	ND	MOHWM0016.02

* MDL = 10 CFU/g.

** MDL = 0.01 ppm.

*** MDL = 3.0 CFU/g.

**** MDL = 100 CFU/g.

CFU – Colony forming units; CNS – Chinese national standard; g – Grams; IDF – International dairy federation; ISO – International organization for standardization; LOQ – Limit of quantification; MOHWM – Taiwan Ministry of Health and Welfare Methods; MDL – Method detection limit; MPN – Most probable number; ND – Not detected; ppm – Parts per million; rRNA – Ribosomal ribonucleic acid; TFU – Total fluorescent units

Conformance with the specifications is demonstrated by the results of 5 nonconsecutive lots, presented in Table 4.

Table 4. Specifications and Analysis of ReviveBio, Co.'s Autoclaved *P. goldsteinii* RV-01

Parameter	Specification	Autoclaved <i>P. goldsteinii</i> RV-01 Representative Lots				
		20231213	20231220	20231226	20240118	20240215
Appearance	Off-white to beige homogenous powder	Pass	Pass	Pass	Pass	Pass
Color	Off-white to beige	Pass	Pass	Pass	Pass	Pass
Odor	Characteristic	Pass	Pass	Pass	Pass	Pass
Moisture (%)	< 5.0	1.5	1.7	1.1	0.8	1.9
Water Activity	< 2.0	0.1	0.04	0.1	0.1	0.1
Identification (%)	> 98% <i>P. goldsteinii</i> RV-01	100	100	100	100	100
<i>P. goldsteinii</i> RV-01 Count (TFU/g)	> 1.2 x 10 ¹⁰	4.24 x 10 ¹⁰	2.13 x 10 ¹⁰	5.48 x 10 ¹⁰	2.91 x 10 ¹⁰	2.78 x 10 ¹⁰
Viable <i>P. goldsteinii</i> Count (CFU/g)*	ND	ND	ND	ND	ND	ND
Heavy Metals						
Arsenic (ppm)	< 0.1	0.05	0.05	0.04	0.05	0.05
Lead (ppm)	< 0.1	0.04	0.04	0.04	0.03	0.04
Cadmium (ppm)	< 0.1	0.05	0.05	0.04	0.04	0.04
Mercury (ppm)**	< 0.1	ND	ND	ND	ND	ND
Microorganisms						
Aerobic Plate Count (CFU/g)*	< 500	ND	ND	ND	ND	ND
Yeast and Molds (CFU/g)*	< 100	ND	ND	ND	ND	ND
<i>Salmonella</i> (per 25 g)	Absent	Absent	Absent	Absent	Absent	Absent
<i>Listeria monocytogenes</i>	ND	ND	ND	ND	ND	ND

Parameter	Specification	Autoclaved <i>P. goldsteinii</i> RV-01 Representative Lots				
		20231213	20231220	20231226	20240118	20240215
(CFU/g)*						
<i>Staphylococcus aureus</i> (CFU/g)*	ND	ND	ND	ND	ND	ND
<i>Escherichia coli</i> (MPN/g)***	ND	ND	ND	ND	ND	ND
<i>Enterobacteriaceae</i> (CFU/g)*	ND	ND	ND	ND	ND	ND
<i>Bacillus cereus</i> (CFU/g)****	ND	ND	ND	ND	ND	ND
<i>C. perfringens</i> (CFU/g)*	N/A	ND	ND	ND	ND	ND

* MDL = 10 CFU/g.

** MDL = 0.01 ppm.

*** MDL = 3.0 CFU/g.

**** MDL = 100 CFU/g.

CFU – Colony forming units; g – Grams; ISO – International Organization for Standardization; LOQ – Limit of quantification; MDL – Method detection limit; MPN – Most probable number; N/A - Not applicable; ND – Not detected; ppm – Parts per million; TFU – Total fluorescent units

A representative analysis of viable *P. goldsteinii* RV-01 cells is provided in Figure 3. In addition, ReviveBio, Co. has analyzed 5 non-consecutive lots for *Clostridium* (*C.*) *perfringens*, a pathogenic and spore-forming bacteria, and the results indicate that the organism is consistently not detected (< 10 CFU/g) in the autoclaved *P. goldsteinii* RV-01 ingredient, using a validated method for *C. perfringens* in foods (MOHWM0006.02).

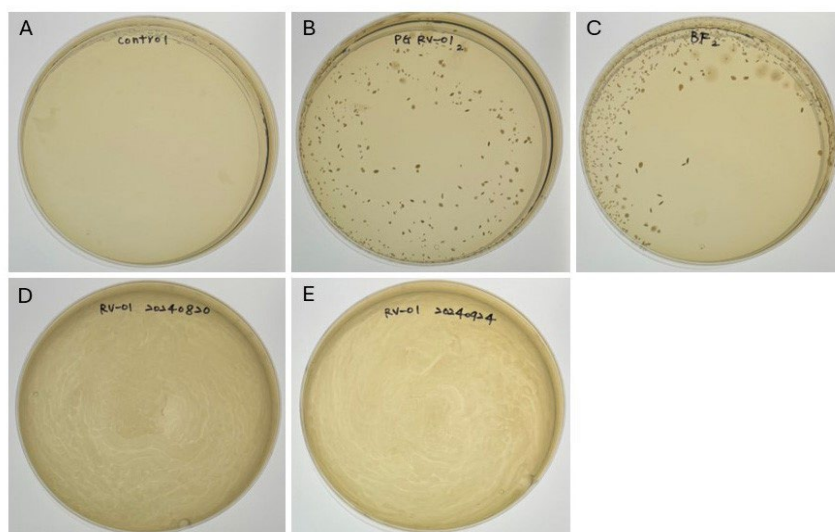


Figure 3. Representative Analysis of Viable *P. goldsteinii* RV-01 Count⁴

⁴ The growth of the diluent (1 mL PBS) (A), viable *P. goldsteinii* RV-01 (100-500 CFU/mL) (B), viable *Bacteroides fragilis* ATCC25285 (100-500 CFU/mL) (C), and two lots of the autoclaved *P. goldsteinii* RV-01 ingredient (D and E; D, Lot 20240820; E, Lot 20240924) on BHIS plates containing kanamycin and vancomycin. Plates D and E were inoculated with 1 mL of a solution containing 0.1 g of ingredient/mL (approximately 5×10^9 TFU). Colonies were observed on the plates with viable *P. goldsteinii* RV-01 (B) and *B. fragilis* ATCC25285 (C). No colonies were detected on the plates with diluent (A) or the autoclaved *P. goldsteinii* RV-01 ingredient (D and E).

In addition to the specifications and analysis provided above, the nutritional profile (based on 5 non-consecutive lots) of autoclaved *P. goldsteinii* RV-01 is provided in Table 5.

Table 5. Typical Nutritional Profile of the Autoclaved *P. goldsteinii* RV-01 Ingredient

Parameter	Typical Range (per 100 g)	Method
Protein (g)	2.5 - 3.0	AOAC 992.15 ¹
Fat (g)	0.4 - 0.9	CNS 5036
Ash (g)	0.2 - 0.3	CNS 5034
Carbohydrate (g)	95.1 - 96.3	Calculated ²
Energy (Kcal)	397.8 - 401.7	Calculated ³

¹This method detects all proteins including those which are denatured.

²Carbohydrate = 100 - (Protein + Fat + Moisture + Ash)

³Energy = [(4 * Protein) + (9 * Fat) + (4 * Carbohydrate)]

AOAC - Association of Official Analytical Collaboration; CNS – Chinese National Standard; g – Grams; Kcal – Kilocalories

F. Stability of Autoclaved *P. goldsteinii* RV-01

ReviveBio, Co. has evaluated the stability of autoclaved *P. goldsteinii* RV-01. The stability of 3 batches of the ingredient was tested in sealed aluminum bags. Stability testing was conducted in climate chambers under 2 different conditions: a temperature of 25°C with a relative humidity of 60% (ambient conditions), and a temperature of 40°C with a relative humidity of 75% (accelerated conditions). The results demonstrate that the product is stable under accelerated conditions for 3 months (Table 6) and under ambient conditions for at least 6 months (Table 7).

Table 6. Autoclaved *P. goldsteinii* RV-01 Accelerated Storage Stability Data

Parameter	Specification	Lot # 20231226		
		0 Months	1 Month	3 Months
Appearance	Off-white to beige homogenous powder	Pass	Pass	Pass
Color	Off-white to beige	Pass	Pass	Pass
Odor	Characteristic	Pass	Pass	Pass
Moisture (%)	< 5.0	1.1	1.3	1.4
Water Activity	< 2.0	0.1	0.1	0.1
Total Cell Count (TFU/g)	> 1.2x10 ¹⁰	5.48 x 10 ¹⁰	5.15 x 10 ¹⁰	5.20 x 10 ¹⁰
Viable Cell Count (CFU/g)	< 10	ND	ND	ND
Aerobic Plate Count (CFU/g)*	< 500	ND	20	ND
Yeast and Molds (CFU/g)*	< 100	ND	ND	ND
<i>Salmonella</i> (per 25 g)	Absent	Absent	Absent	Absent
<i>Listeria monocytogenes</i> (CFU/g)*	< 10	ND	ND	ND
<i>Staphylococcus aureus</i> (CFU/g)*	≤ 10	ND	ND	ND
<i>Escherichia coli</i> (MPN/g)**	< 3.0	ND	ND	ND
<i>Enterobacteriaceae</i> (CFU/g)*	≤ 10	ND	ND	ND
<i>Bacillus cereus</i> (CFU/g)***	≤ 100	ND	ND	ND
Arsenic (ppm)	< 0.1	0.04	0.04	0.05
Lead (ppm)	< 0.1	0.04	0.04	0.04
Cadmium (ppm)	< 0.1	0.04	0.04	0.04
Mercury (ppm)****	< 0.1	ND	ND	ND
Parameter	Specification	Lot # 20240118		

		0 Months	1 Month	3 Months
Appearance	Off-white to beige homogenous powder	Pass	Pass	Pass
Color	Off-white to beige	Pass	Pass	Pass
Odor	Characteristic	Pass	Pass	Pass
Moisture (%)	< 5.0	0.8	1.7	1.9
Water Activity	< 2.0	0.1	0.1	0.1
Total Cell Count (TFU/g)	> 1.2 x 10 ¹⁰	2.91 x 10 ¹⁰	2.72 x 10 ¹⁰	2.10 x 10 ¹⁰
Viable Cell Count (CFU/g)	< 10	ND	ND	ND
Aerobic Plate Count (CFU/g)*	< 500	ND	ND	ND
Yeast and Molds (CFU/g)*	< 100	ND	ND	ND
Salmonella (per 25 g)	Absent	Absent	Absent	Absent
Listeria monocytogenes (CFU/g)*	< 10	ND	ND	ND
Staphylococcus aureus (CFU/g)*	≤ 10	ND	ND	ND
Escherichia coli (MPN/g)**	< 3.0	ND	ND	ND
Enterobacteriaceae (CFU/g)*	≤ 10	ND	ND	ND
Bacillus cereus (CFU/g)***	≤ 100	ND	ND	ND
Arsenic (ppm)	< 0.1	0.05	0.05	0.05
Lead (ppm)	< 0.1	0.03	0.04	0.03
Cadmium (ppm)	< 0.1	0.04	0.04	0.04
Mercury (ppm)****	< 0.1	ND	ND	ND
Parameter	Specification	Lot # 20240215		
		0 Months	1 Month	3 Months
Appearance	Off-white to beige homogenous powder	Pass	Pass	Pass
Color	Off-white to beige	Pass	Pass	Pass
Odor	Characteristic	Pass	Pass	Pass
Moisture (%)	< 5.0	1.9	1.8	2.1
Water Activity	< 2.0	0.1	0.1	0.1
Total Cell Count (TFU/g)	> 1.2 x 10 ¹⁰	2.78 x 10 ¹⁰	2.36 x 10 ¹⁰	2.84 x 10 ¹⁰
Viable Cell Count (CFU/g)*	< 10	ND	ND	ND
Aerobic Plate Count (CFU/g)*	< 500	ND	ND	ND

Yeast and Molds (CFU/g)*	< 100	ND	ND	ND
<i>Salmonella</i> (per 25 g)	Absent	Absent	Absent	Absent
<i>Listeria monocytogenes</i> (CFU/g)*	< 10	ND	ND	ND
<i>Staphylococcus aureus</i> (CFU/g)*	≤ 10	ND	ND	ND
<i>Escherichia coli</i> (MPN/g)**	< 3.0	ND	ND	ND
<i>Enterobacteriaceae</i> (CFU/g)*	≤ 10	ND	ND	ND
<i>Bacillus cereus</i> (CFU/g)***	≤ 100	ND	ND	ND
Arsenic (ppm)	< 0.1	0.05	0.05	0.05
Lead (ppm)	< 0.1	0.04	0.04	0.03
Cadmium (ppm)	< 0.1	0.04	0.04	0.04
Mercury (ppm)****	< 0.1	ND	ND	ND

*MDL = 10 CFU/g.

**MDL = 3.0 MPN/g.

***MDL = 100 CFU/g.

****MDL = 0.01 ppm.

CFU – Colony forming units; g – Grams; MDL – Minimum detection limit; MPN – Most probable number; ND – Not detected; ppm – Parts per million; TFU – Total fluorescent units

Table 7. Autoclaved *P. goldsteinii* RV-01 Ambient Storage Stability Data

Parameter	Specification	Lot # 20231226		
		0 Months	3 Months	6 Months
Appearance	Off-white to beige homogenous powder	Pass	Pass	Pass
Color	Off-white to beige	Pass	Pass	Pass
Odor	Characteristic	Pass	Pass	Pass
Moisture (%)	< 5.0	1.1	1.4	1.8
Water Activity	< 2.0	0.1	0.1	0.1
Total Cell Count (TFU/g)	> 1.2 x 10 ¹⁰	5.48 x 10 ¹⁰	5.20 x 10 ¹⁰	5.39 x 10 ¹⁰
Viable Cell Count (CFU/g)	< 10	ND	ND	ND
Aerobic Plate Count (CFU/g)*	< 500	ND	ND	ND
Yeast and Molds (CFU/g)*	< 100	ND	ND	ND
<i>Salmonella</i> (per 25 g)	Absent	Absent	Absent	Absent
<i>Listeria monocytogenes</i> (CFU/g)*	< 10	ND	ND	ND

<i>Staphylococcus aureus</i> (CFU/g)*	≤ 10	ND	ND	ND
<i>Escherichia coli</i> (MPN/g)**	< 3.0	ND	ND	ND
<i>Enterobacteriaceae</i> (CFU/g)*	≤ 10	ND	ND	ND
<i>Bacillus cereus</i> (CFU/g)***	≤ 100	ND	ND	ND
Arsenic (ppm)	< 0.1	0.04	0.05	0.04
Lead (ppm)	< 0.1	0.04	0.04	0.03
Cadmium (ppm)	< 0.1	0.04	0.04	0.04
Mercury (ppm)****	< 0.1	ND	ND	ND
Parameter	Specification	Lot # 20240118		
		0 Months	3 Months	6 Months
Appearance	Off-white to beige homogenous powder	Pass	Pass	Pass
Color	Off-white to beige	Pass	Pass	Pass
Odor	Characteristic	Pass	Pass	Pass
Moisture (%)	< 5.0	0.8	1.5	1.4
Water Activity	< 2.0	0.1	0.1	0.1
Total Cell Count (TFU/g)	> 1.2 x 10 ¹⁰	2.91 x 10 ¹⁰	2.19 x 10 ¹⁰	2.40 x 10 ¹⁰
Viable Cell Count (CFU/g)	< 10	ND	ND	ND
Aerobic Plate Count (CFU/g)*	< 500	ND	ND	ND
Yeast and Molds (CFU/g)*	< 100	ND	ND	ND
<i>Salmonella</i> (per 25 g)	Absent	Absent	Absent	Absent
<i>Listeria monocytogenes</i> (CFU/g)*	< 10	ND	ND	ND
<i>Staphylococcus aureus</i> (CFU/g)*	≤ 10	ND	ND	ND
<i>Escherichia coli</i> (MPN/g)**	< 3.0	ND	ND	ND
<i>Enterobacteriaceae</i> (CFU/g)*	≤ 10	ND	ND	ND
<i>Bacillus cereus</i> (CFU/g)***	≤ 100	ND	ND	ND
Arsenic (ppm)	< 0.1	0.05	0.05	0.05
Lead (ppm)	< 0.1	0.03	0.03	0.03
Cadmium (ppm)	< 0.1	0.04	0.04	0.04
Mercury (ppm)****	< 0.1	ND	ND	ND
Parameter	Specification	Lot # 20240215		

		0 Months	3 Months	6 Months
Appearance	Off-white to beige homogenous powder	Pass	Pass	Pass
Color	Off-white to beige	Pass	Pass	Pass
Odor	Characteristic	Pass	Pass	Pass
Moisture (%)	< 5.0	1.9	1.4	1.4
Water Activity	< 2.0	0.1	0.1	0.1
Total Cell Count (TFU/g)	> 1.2 x 10 ¹⁰	2.78 x 10 ¹⁰	2.56 x 10 ¹⁰	2.64 x 10 ¹⁰
Viable Cell Count (CFU/g)*	< 10	ND	ND	ND
Aerobic Plate Count (CFU/g)*	< 500	ND	ND	ND
Yeast and Molds (CFU/g)*	< 100	ND	ND	ND
Salmonella (per 25 g)	Absent	Absent	Absent	Absent
Listeria monocytogenes (CFU/g)*	< 10	ND	ND	ND
Staphylococcus aureus (CFU/g)*	≤ 10	ND	ND	ND
Escherichia coli (MPN/g)**	< 3.0	ND	ND	ND
Enterobacteriaceae (CFU/g)*	≤ 10	ND	ND	ND
Bacillus cereus (CFU/g)***	≤ 100	ND	ND	ND
Arsenic (ppm)	< 0.1	0.05	0.04	0.04
Lead (ppm)	< 0.1	0.04	0.03	0.04
Cadmium (ppm)	< 0.1	0.04	0.04	0.03
Mercury (ppm)****	< 0.1	ND	ND	ND

*MDL = 10 CFU/g.

**MDL = 3.0 MPN/g.

***MDL = 100 CFU/g.

****MDL = 0.01 ppm.

CFU – Colony forming units; g – Grams; MDL – Minimum detection limit; MPN – Most probable number; N/A - Not available; ND – Not detected; ppm – Parts per million; TFU – Total fluorescent units

PART 3. DIETARY EXPOSURE

ReviveBio, Co.'s autoclaved *P. goldsteinii* RV-01 is intended to be used as an ingredient in the following conventional food categories: sports and energy drinks, ready-to-eat cereals (low sugar), yogurt, fermented milk products, ice cream, cereal bars, nutrition bars, nut spreads, and candy containing chocolate. The intended use of the ingredient is up to 1×10^{10} TFU/serving, or 185 mg/serving. The intended use does not include infant formula or USDA-regulated foods.

A. Estimate of Dietary Exposure to the Substance

The proposed food categories and inclusion levels are presented in Table 8. It is noted that some food categories include standardized foods. The autoclaved *P. goldsteinii* RV-01 ingredient will be added to a standardized food only if permitted by the applicable standard of identity.

Table 8. Intended Food Categories and Use Levels of Autoclaved *P. goldsteinii* RV-01

Food Categories	Use Level (TFU/serving)	RACC ¹ (g or mL)	Use Level (TFU/g or TFU/mL)
Sports and Energy Drinks	1×10^{10}	360	2.8×10^7
Ready-to-eat cereal (low sugar)	1×10^{10}	15	6.7×10^8
Yogurt	1×10^{10}	170	5.9×10^7
Fermented Milk Products	1×10^{10}	240	4.2×10^7
Ice Cream	1×10^{10}	160	6.3×10^7
Cereal and Nutrition Bars	1×10^{10}	40	2.5×10^8
Nut Spreads	1×10^{10}	32	3.1×10^8
Candy Containing Chocolate	1×10^{10}	30	3.3×10^8

¹ Serving size determined from Reference Amount Customarily Consumed (RACC) - 21 CFR § 101.12. Assumption that 1 g = 1 mL for RACC.

g- Grams; mL- Milliliters; TFU – Total fluorescent units

The estimated dietary exposures for autoclaved *P. goldsteinii* RV-01 were calculated using the Creme Food Safety® model (<https://www.cremeglobal.com>). The model includes food consumption data included in the NHANES 2017-2020 pre-pandemic survey (CDC, 2020). Output calculation types include daily average intakes, acute exposures, as well as population statistics such as mean, percentiles, standard errors, and confidence intervals. Results are output for “Consumers Only” (i.e., consumers of the intended food categories). Results of the exposure assessment are given in absolute terms (TFU/day) as well as relative to the consumer’s body weight (TFU/kg body weight (bw)/day) at the mean and 90th percentile. No evidence has identified *P. goldsteinii* RV-01 as being found naturally in the diet, so the consumption estimate is calculated based on the addition of autoclaved *P. goldsteinii* RV-01 to specific foods (Table 8). To calculate the amount of TFU, 1 mg of autoclaved *P. goldsteinii* RV-01 = 54,060,000 TFU. This calculation was based on the Lin et al.

(2024) publication in which 1,500 mg/kg bw/day was reported to be equal to 8.109×10^{10} TFU/kg bw/day.

The per unit of body weight exposure is calculated on a subject level using the body weight recorded by the NHANES data. The results for “Consumers Only” are shown in Table 9 and Table 10 and a detailed report is provided in Appendix 6. The intended use is not in infant formula or foods for young children; therefore, intake estimates for 0–1-year-olds are not included.

Table 9. Summary of Consumer Only Intake (mg) from the Proposed Use of Autoclaved *P. goldsteinii* RV-01 Based on NHANES 2017-2020 Survey

Population Group	N	% Pop.	Consumer Only Intake (mg/day)		Consumer Only Intake (mg/kg bw/day)	
			Mean ± SE	90 th Percentile ± SE	Mean ± SE	90 th Percentile ± SE
Children, Females (1-5 yrs)	331	66%	147.0 ± 6.6	277.5 ± 20.1	9.3 ± 0.5	18.2 ± 1.3
Children, Males (1-5 yrs)	339	62%	158.9 ± 8.3	342.7 ± 30.5	9.8 ± 0.6	19.8 ± 1.7
Children, Females (6-11 yrs)	408	68%	186.3 ± 6.8	375.3 ± 19.2	5.6 ± 0.2	11.9 ± 0.6
Children, Males (6-11 yrs)	366	61%	176.8 ± 7.6	403.6 ± 32.7	5.6 ± 0.3	13.1 ± 0.9
Teenage, Female (12-18 yrs)	372	57%	246.5 ± 9.8	581.1 ± 46.7	4.2 ± 0.2	10.4 ± 0.7
Teenage, Male (12-18 yrs)	360	54%	238.7 ± 12.5	513.9 ± 43.4	3.7 ± 0.2	8.2 ± 0.5
Adults, Female (19-110 yrs)	2011	57%	190.7 ± 3.5	402.5 ± 11.8	2.7 ± 0.1	5.9 ± 0.1
Adults, Male (19-110 yrs)	1743	54%	279.1 ± 7.1	618.5 ± 23.4	3.2 ± 0.1	7.1 ± 0.3
Males and Females (1-110 yrs)	5930	57%	223.8 ± 2.5	475.9 ± 5.4	3.4 ± 0.1	8.2 ± 0.2

bw – Body weight; kg – Kilograms; mg – Milligrams; Pop – Population; SE – Standard error; yrs – Years

Table 10. Summary of Consumer Only Intake (TFU) from the Proposed Use of Autoclaved *P. goldsteinii* RV-01 Based on NHANES 2017-2020 Survey

Population Group	N	% Pop.	Consumer Only Intake (TFU/day)		Consumer Only Intake (TFU/kg bw/day)	
			Mean ± SE	90 th Percentile ± SE	Mean ± SE	90 th Percentile ± SE
Children, Females (1-5 yrs)	331	66%	7.9×10^9 ($\pm 3.5 \times 10^8$)	1.5×10^{10} ($\pm 1.1 \times 10^9$)	5.1×10^8 ($\pm 2.6 \times 10^7$)	9.8×10^8 ($\pm 7.0 \times 10^7$)
Children, Males (1-5 yrs)	339	62%	8.6×10^9 ($\pm 4.5 \times 10^8$)	1.9×10^{10} ($\pm 1.6 \times 10^9$)	5.3×10^8 ($\pm 3.0 \times 10^7$)	1.1×10^9 ($\pm 9.2 \times 10^7$)
Children, Females (6-11 yrs)	408	68%	1.0×10^{10} ($\pm 3.7 \times 10^8$)	2.0×10^{10} ($\pm 1.0 \times 10^9$)	3.0×10^8 ($\pm 1.2 \times 10^7$)	6.5×10^8 ($\pm 3.4 \times 10^7$)
Children, Males (6-11 yrs)	366	61%	9.6×10^9 ($\pm 4.1 \times 10^8$)	2.2×10^{10} ($\pm 1.8 \times 10^9$)	3.0×10^8 ($\pm 1.6 \times 10^7$)	7.1×10^8 ($\pm 4.6 \times 10^7$)
Teenage, Female (12-18 yrs)	372	57%	1.3×10^{10} ($\pm 5.3 \times 10^8$)	3.1×10^{10} ($\pm 2.5 \times 10^9$)	2.3×10^8 ($\pm 9.9 \times 10^6$)	5.6×10^8 ($\pm 4.0 \times 10^7$)
Teenage, Male (12-18 yrs)	360	54%	1.3×10^{10} ($\pm 6.8 \times 10^8$)	2.8×10^{10} ($\pm 2.3 \times 10^9$)	2.0×10^8 ($\pm 9.7 \times 10^6$)	4.4×10^8 ($\pm 2.8 \times 10^7$)

Population Group	N	% Pop.	Consumer Only Intake (TFU/day)		Consumer Only Intake (TFU/kg bw/day)	
			Mean $\pm$ SE	90 th Percentile $\pm$ SE	Mean $\pm$ SE	90 th Percentile $\pm$ SE
Adults, Female (19-110 yrs)	2011	57%	1.0×10^{10} ($\pm 1.9 \times 10^8$)	2.2×10^{10} ($\pm 6.4 \times 10^8$)	1.4×10^8 ($\pm 2.8 \times 10^6$)	3.2×10^8 ($\pm 5.7 \times 10^6$)
Adults, Male (19-110 yrs)	1743	54%	1.5×10^{10} ($\pm 3.8 \times 10^8$)	3.3×10^{10} ($\pm 1.3 \times 10^9$)	1.7×10^8 ($\pm 3.9 \times 10^6$)	3.9×10^8 ($\pm 1.4 \times 10^7$)
Males and Females (1-110 yrs)	5930	57%	1.2×10^{10} ($\pm 1.4 \times 10^8$)	2.6×10^{10} ($\pm 2.9 \times 10^8$)	2.0×10^8 ($\pm 3.2 \times 10^6$)	4.4×10^8 ($\pm 1.0 \times 10^7$)

bw – Body weight; kg – Kilograms; Pop – Population; SE – Standard error; TFU – Total Fluorescent Units; yrs – Years

Estimated mean and 90th percentile consumption of autoclaved *P. goldsteinii* RV-01 for males and females of all ages is 223.8 ± 2.5 mg/day (3.4 ± 0.1 mg/kg bw/day) and 475.9 ± 5.4 mg/day (8.2 ± 0.2 mg/kg bw/day), respectively. In the population with the highest absolute consumption (adult males), the mean and 90th percentile intakes at the proposed maximum intended use of autoclaved *P. goldsteinii* RV-01 are 279.1 ± 7.1 mg/day (3.4 ± 0.1 mg/kg bw/day) and 618.5 ± 23.4 mg/day (7.1 ± 0.3 mg/kg bw/day), respectively. In terms of intake relative to body weight, male children aged 1-5 years have the highest estimated intake of autoclaved *P. goldsteinii* RV-01 at the mean and 90th percentile; 9.8 ± 0.6 mg/kg bw/day and 19.8 ± 1.7 mg/kg bw/day, respectively.

Estimated mean and 90th percentile consumption of autoclaved *P. goldsteinii* RV-01 on a TFU basis for all ages is 1.2×10^{10} TFU/day (2.0×10^8 TFU/kg bw/day) and 2.6×10^{10} TFU/day (4.4×10^8 TFU/kg bw/day), respectively. In the population with the highest absolute consumption (adult males), the mean and 90th percentile intakes at the proposed maximum intended use of autoclaved *P. goldsteinii* RV-01 are 1.5×10^{10} TFU/day (1.7×10^8 TFU/kg bw/day) and 3.3×10^{10} TFU/day (3.9×10^8 TFU/kg bw/day), respectively. In terms of intake relative to body weight, male children aged 1-5 years have the highest estimated intake of autoclaved *P. goldsteinii* RV-01 at the mean and 90th percentile; 5.3×10^8 TFU/kg bw/day and 1.1×10^9 TFU/kg bw/day, respectively. However, this is due to the lower bodyweight of this subpopulation, as the absolute intake of autoclaved *P. goldsteinii* RV-01 in male children aged 1-5 years (1.9×10^{10} TFU/day) is about 58% of the intake in adult males (3.3×10^{10} TFU/day) and 73% of the intake in the general population (2.6×10^{10} TFU/day).

B. Estimated Dietary Exposure to Any Other Substance That is Expected to be Formed in or On Food

This section is not relevant to autoclaved *P. goldsteinii* RV-01, as the ingredient is expected to be stable under the intended conditions of use.

C. Dietary Exposure to Contaminants or Byproducts

Potential contaminants in ReviveBio, Co.'s autoclaved *P. goldsteinii* RV-01 include microbes and heavy metals. The specifications set for ReviveBio, Co.'s autoclaved *P. goldsteinii* RV-01 place limits on the maximum permissible levels of these contaminants to ensure an acceptable final product. The batch data for five nonconsecutive lots of autoclaved *P. goldsteinii* RV-01 document the quality

control of the ingredient such that it consistently meets these specifications (Table 4). Maltodextrin may contain copper at a concentration up to 5 parts per million (ppm), but the maltodextrin content of the autoclaved *P. goldsteinii* RV-01 ingredient is about 85.7% (maltodextrin is added at a ratio of 6:1 by weight of autoclaved *P. goldsteinii* RV-01). The tolerable upper limit (UL) for copper intake varies across the lifespan, from 1 mg/day in children aged 1-3 years, up to 10 mg/day in adults (Oregon State University, 2024). The highest 90th percentile intakes of autoclaved *P. goldsteinii* RV-01 in children aged 1-5 years and adults are 342.7 and 618.5 mg/day, respectively. Assuming the highest permissible level of copper is present in the maltodextrin (5 ppm), and the maltodextrin content of the ingredient (85.7%), the intake of copper from 90th percentile intake of the autoclaved *P. goldsteinii* RV-01 ingredient is up to 0.15 and 0.27 mg/day in children aged 1-5 years and adults, respectively. These intakes represent 15% and 2.7% of the ULs for copper in young children and adults, respectively. Thus, the presence of copper in the autoclaved *P. goldsteinii* RV-01 ingredient is not expected to pose a significant risk of toxicity.

In terms of microbial contamination, the specification for *Listeria monocytogenes* established by ReviveBio, Co. (< 10 CFU/g) is lower than the limit established by EFSA for ready-to-eat foods (excluding foods to be consumed by infants) of < 100 CFU/g, according to European Commission (EC) No 2073/2005.⁵ The results of 5 non-consecutive lots presented in Table 4 indicate that *Listeria monocytogenes* is consistently not detected in autoclaved *P. goldsteinii* RV-01. Further, there is no established limit for *Bacillus (B.) cereus* in ready-to-eat foods according to the applicable European regulations. In infant formula, the limit is 50 CFU/g, which is similar to the specification for *B. cereus* in autoclaved *P. goldsteinii* RV-01 (< 100 CFU/g), although the intended use of ReviveBio, Co's autoclaved *P. goldsteinii* RV-01 does not include infant formula. According to EFSA (2016), "most cases of food-borne outbreaks caused by the *B. cereus* group have been associated with bacterial concentrations above 10⁵ CFU/g of foodstuff. However, it is important to highlight that cases of both emetic and diarrheal illness have been reported involving between 10³ and 10⁵ CFU/g". This concentration is at least 10-1,000 times greater than the maximum level of *B. cereus* permitted in the autoclaved *P. goldsteinii* RV-01 ingredient (< 100 CFU/g). Importantly, the results of 5 non-consecutive lots presented in Table 4 indicate that *B. cereus* is consistently not detected in autoclaved *P. goldsteinii* RV-01.

During the growth of live *P. goldsteinii* RV-01, short chain fatty acids (SCFAs) are produced from carbohydrates. Given the presence of carbohydrates in the final ingredient (Table 5), 7 representative lots of autoclaved *P. goldsteinii* RV-01 have been analyzed using liquid chromatography-mass spectrometry (LC-MS) for several SCFAs. The results of this analysis are presented in Table 11.

The primary short chain fatty acid in autoclaved *P. goldsteinii* RV-01 is acetic acid, at a concentration of up to about 11,900 nmol/g, or 0.0000119 mol/g, which is equivalent to 0.00071 g/g or 0.71 mg/g of the ingredient.⁶ At the highest 90th percentile intake of autoclaved *P. goldsteinii* RV-01 (618.5

⁵ Accessed on April 22, 2025, from: [EUR-Lex - 02005R2073-20200308 - EN - EUR-Lex](#)

⁶ The molar mass of acetic acid is 60.05 g/mol ([Acetic Acid | CH3COOH | CID 176 - PubChem \(nih.gov\)](#))

mg/day) the intake of acetic acid would be 0.44 mg/day. Acetic acid/acetate is a primary byproduct of microbial fermentation in the human gut, typically derived from fibers such as galactooligosaccharides and inulin and typically the most highly concentrated SCFA in the human gut lumen (Facchin et al., 2024). In addition, Valdes et al. (2021) conducted a meta-analysis of clinical trials involving acetic acid supplementation at doses of 750 to 3,600 mg/day for up to 8 weeks. A total of 16 studies were included, involving 910 participants, and the authors ultimately concluded that, “supplementation with dietary acetic acid is well tolerated, [and] has no adverse side effects”. Thus, it is reasonable to assume that the presence of acetic acid in the autoclaved *P. goldsteinii* RV-01 ingredient will not adversely affect its safety.

Another SCFA present in autoclaved *P. goldsteinii* RV-01 is propionic acid, at a concentration of up to about 550 nmol/g, or 0.041 mg/g of the ingredient.⁷ At the highest 90th percentile intake of autoclaved *P. goldsteinii* RV-01 (618.5 mg/day in adult males), the intake of propionic acid would be about 0.025 mg/day, or 0.36 µg/kg bw/day in a 70 kg adult. In children aged 1-5 years, the highest 90th percentile intake of autoclaved *P. goldsteinii* RV-01 is 342.7 mg/day, corresponding to an intake of propionic acid at 0.014 mg/day, or 0.81 µg/kg bw/day using the bodyweight of 1-5-year-old males from NHANES (17.31 kg).⁸ Propionic acid/propionate is one of the primary SCFAs (along with acetate and butyrate) produced by the human intestinal flora via fermentation of plant fibers such as cellulose, pectin, hemicellulose, and lignin (Markowiak-Kopec and Slizewska, 2020). In addition, EFSA (2016) published a scientific opinion on the safety of propionic acid and its major salts (sodium, calcium, and potassium propionate) as food additives. The panel estimated that 95th percentile intake of propionic acid and propionate salts from food is 3.6 – 40.8 mg/kg bw/day. While an acceptable daily intake (ADI) is not currently established⁹, the panel concluded that there is no safety concern at the 95th percentile intake. This level of intake is 10,000 – 113,333 times greater than the 90th percentile intake of propionic acid from autoclaved *P. goldsteinii* RV-01 in adult males and 4,444 – 50,370 times greater than the 90th percentile intake of propionic acid from autoclaved *P. goldsteinii* RV-01 in male children aged 1-5 years. Therefore, the presence of propionic acid in the autoclaved *P. goldsteinii* RV-01 ingredient is not expected to adversely affect its safety.

Formic acid is also present in autoclaved *P. goldsteinii* RV-01 and represents a potential safety concern given its irritating properties when ingested (CDC, 2024). This substance can also cause metabolic acidosis.¹⁰ The concentration of formic acid in autoclaved *P. goldsteinii* RV-01 is up to about 7,900 nmol/g, which corresponds to 0.36 mg/g of the ingredient.¹¹ According to the Joint FAO/WHO Expert Committee on Food Additives (JECFA), the ADI of formic acid is up to 3 mg/kg bw/day (WHO, 1997). Using the highest 90th percentile intake of autoclaved *P. goldsteinii* RV-01 (618.5 mg/day in adult males), the highest estimated daily intake of formic acid is about 0.22 mg/day or 3.2 µg/kg bw/day assuming a 70 kg bodyweight, which is about 1,000 times less than the ADI of 3

⁷ The molar mass of propionic acid is 74.08 g/mol ([Propionic Acid | CH3CH2COOH | CID 1032 - PubChem](#))

⁸ The bodyweight was calculated by dividing the 90th percentile intake in mg/day by the 90th percentile intake in mg/kg bw/day from Table 9.

⁹ Accessed on April 22, 2025, from: [WHO | JECFA](#)

¹⁰ Accessed on April 22, 2025, from: [Methanol Toxicity - StatPearls - NCBI Bookshelf](#)

¹¹ The molar mass of formic acid is 46.0 g/mol ([CDC - NIOSH Pocket Guide to Chemical Hazards - Formic acid](#))

mg/kg bw/day. In children aged 1-5 years, the highest 90th percentile intake of autoclaved *P. goldsteinii* RV-01 is 342.7 mg/day (in males), corresponding to an intake of formic acid of 0.12 mg/day, or 7.1 µg/kg bw/day, which is 423 times less than the ADI. Thus, the presence of formic acid in autoclaved *P. goldsteinii* RV-01 is not expected to adversely affect its safety. Further, a statistical analysis of the results using the method described by Motulsky and Brown (2006) indicates that the highest reported value for formic acid (7,877.39 nmol/g) is an outlier and thus the typical formic acid content is expected to be about 6.6 times lower, based on the mean of the other 6 lots; 1,192 nmol/g.

Table 11. Short Chain Fatty Acids in Autoclaved *P. goldsteinii* RV-01

Parameter (nmol/g)	Autoclaved <i>P. goldsteinii</i> RV-01 Representative Lots						
	20231226	20240118	20240215	20231218	20231220	20231222	20240103
Formic acid	1,160.81	1,178.90	7,877.39	1,301.20	1,185.23	1,088.73	1,237.00
Acetic acid	10,274.54	10,568.33	11,898.98	9,039.49	10,055.27	10,114.28	10,302.63
Propionic acid	406.31	434.10	548.04	298.53	338.72	305.22	379.92
Butyric acid	184.01	188.86	268.80	154.08	168.22	145.72	156.23
Isobutyric acid	60.75	80.63	69.58	65.06	49.88	51.87	75.75
Pentanoic acid (valeric acid)	ND	ND	2.87	1.67	1.30	ND	1.32
3-Methylbutyric acid (isovaleric acid)	111.23	81.49	90.62	77.25	106.35	108.29	78.74
2-Methylbutyric acid	83.43	105.47	87.09	89.22	74.00	72.91	93.85
3-Methylpentanoic acid	ND	ND	ND	ND	ND	ND	ND
4-Methylpentanoic acid	1.05	0.61	3.43	ND	ND	0.52	ND
Hexanoic acid	3.17	5.20	5.93	5.45	5.70	3.80	5.40

g – Grams; ND – Not detected; nmol – Nanomole

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PART 4. SELF-LIMITING LEVELS OF USE

No known self-limiting levels of use are associated with autoclaved *P. goldsteinii* RV-01.

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PART 5. EXPERIENCE BASED ON COMMON USE IN FOOD BEFORE 1958

The statutory basis for the conclusion of GRAS status of autoclaved *P. goldsteinii* RV-01 in this document is based on scientific procedures in accordance with 21 CFR 170.30(a)(b). Therefore, experience based on common use in food before 1958 does not apply. A discussion of the history of safe consumption of autoclaved *P. goldsteinii* RV-01 is discussed in Part 6.

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PART 6. NARRATIVE

A. Introduction

A review of the scientific and regulatory literature related to the use of *P. goldsteinii* in foods was conducted through February 10, 2025. Various government databases were used to search for applicable regulations across jurisdictions. To identify safety studies, literature searches were conducted in PubMed using the terms; [("*Parabacteroides goldsteinii*" or "*P. goldsteinii*") and safety] and ["*Parabacteroides goldsteinii*"].

The safety evaluation of autoclaved *P. goldsteinii* RV-01 is conducted in a manner consistent with up-to-date guidance documents that outline best practices and/or specific jurisdictional requirements for the safety assessment of microorganisms intended for use in food. The guidance documents and expert reviews from regulatory authorities and industry specialists that were utilized in the safety evaluation of autoclaved *P. goldsteinii* RV-01 include the EFSA Qualified Presumption of Safety guidelines (Barlow et al., 2007), the Guidelines for the Evaluation of Probiotics in Food (JECFA, 2002) issued by the Joint FAO/WHO Expert Consultation on Evaluation of Health and Nutritional Properties of Probiotics in Food, and the safety decision tree for evaluating microbial cultures intended for human and animal consumption published by Pariza et al. (2015).

Parabacteroides goldsteinii strains were originally isolated and identified by Song et al. (2005) from intestinal samples collected from 7 human patients with infections¹² (4 with appendicitis, 1 with peritonitis, and 2 with abdominal abscesses). However, *Parabacteroides goldsteinii* is actually a human intestinal tract commensal known to contribute to metabolic regulation (Wu et al., 2019; Cui et al., 2022). *Parabacteroides goldsteinii* and related species of the *Parabacteroides* genus have a relative abundance of 1.27% in the feces. The species are obligate anaerobes that are important contributors to the metabolism of simple sugars and are part of complex microbial populations that form inter-organism nutritional webs involved in cross-feeding interactions with other gut microbes (Cui et al., 2022). Further, production of SCFAs are important metabolic functions of these species (Cui et al., 2022). The species has also been shown to assist in the maturation and development of the immune system in mice (Cervantes-Barragan and Colonna, 2022). Further mouse studies have shown that *P. goldsteinii* produces beta-hexosaminidase, a conserved enzyme identified in the Bacteroidota phylum, which promotes the development of a subtype of CD4+ effector T cell that regulates adaptive immunity in the intestinal mucosa, thereby playing an important role in maintaining homeostasis (Bousbaine et al., 2022).

The *P. goldsteinii* RV-01 strain was originally isolated from the feces of a healthy adult (Lin et al., 2024). To circumvent any possibility that live *P. goldsteinii* RV-01 may cause adverse effects in

¹² *P. goldsteinii* was reported to be of intestinal origin and was isolated from the infected tissues following cases of diverticulitis and necrotic ileum. In each case they were isolated from mixed cultures and were not recovered from blood cultures of these patients.

humans, the strain is killed by autoclaving (15 minutes at 121°C), a commonly used heating procedure (Bharti et al., 2022).

B. History of Safe Consumption

The literature searches described in Part 6.A, as well as several internet searches, did not identify any publications which describe the historical use in food of *P. goldsteinii*, including live or autoclaved *P. goldsteinii* RV-01.

C. Regulatory History

1. United States

A search of FDA's GRAS Notice Inventory website¹³ using the search terms "Parabacteroides goldsteinii", "Parabacteroides", and "goldsteinii" did not return any notifications.

A search of FDA's New Dietary Ingredient Notification (NDIN) list¹⁴ using the search terms "Parabacteroides goldsteinii", "Parabacteroides", and "goldsteinii" did not return any notifications.

A search of the Dietary Supplement and Labeling Database (DSLD)¹⁵ using the search terms "Parabacteroides goldsteinii", "Parabacteroides", and "goldsteinii" did not return any results.

2. Europe

Parabacteroides goldsteinii does not have a QPS designation from EFSA. Furthermore, a search of the EFSA webpage¹⁶ using the terms "Parabacteroides goldsteinii", "Parabacteroides", and "goldsteinii" did not return any results.

3. Canada

A search of Health Canada's natural health product ingredient database¹⁷ using the terms "Parabacteroides goldsteinii", "Parabacteroides", and "goldsteinii" did not return any results. Furthermore, *P. goldsteinii* is not listed in Health Canada's probiotic monograph (Health Canada, 2023).

¹³ Accessed April 22, 2025: [GRAS Notices](#)

¹⁴ Accessed April, 2025: <https://www.fda.gov/food/new-dietary-ingredients-ndi-notification-process/submitted-75-day-premarket-notifications-new-dietary-ingredients>

¹⁵ Accessed on April 22, 2025: [Home | Dietary Supplement Label Database \(DSLD\) | NIH Office of Dietary Supplements](#)

¹⁶ Accessed on April 22, 2025: [EFSA | Science, safe food, sustainability \(europa.eu\)](#)

¹⁷ Accessed on April 22, 2025: [Natural health products ingredients database \(hc-sc.gc.ca\)](#)

4. Australia and New Zealand

A search of the Food Standards Australia and New Zealand (FSANZ) webpage¹⁸ using the terms “*Parabacteroides goldsteinii*”, “*Parabacteroides*”, and “*goldsteinii*” did not return any results.

D. Absorption, Distribution, Metabolism & Excretion (ADME)

It is expected that oral consumption of autoclaved *P. goldsteinii* RV-01 will traverse the intestine for ultimate excretion in the feces. While infection with live *P. goldsteinii* has been reported on rare occasions in humans, none of the cases involved oral consumption of live or autoclaved *P. goldsteinii*; rather, each case resulted from the individuals suffering various injuries or immunocompromising conditions (Awadel-Kariem et al., 2010; Ricco et al., 2022; Peng et al., 2024). Further, the specifications for autoclaved *P. goldsteinii* RV-01 limit the live *P. goldsteinii* RV-01 cell count to < 10 CFU/g, significantly lowering the potential risk of infection.

E. Safety Studies

1. *In silico* and *in vitro* Studies

a. Antimicrobial Resistance

An analysis of the *P. goldsteinii* RV-01 genome was conducted to screen for potential antimicrobial resistance genes using the homology search tools by comparison with the Comprehensive Antibiotic Resistance Database (CARD). To identify antimicrobial resistance sequence matches, thresholds of > 60% coverage and > 70% identity were applied based on relevant EFSA guidance (EFSA, 2021). Only 1 gene was a significant match to the tetracycline resistance gene (*tetQ*) and met the threshold qualifications. However, polymerase chain reaction (PCR) followed by gel electrophoresis indicates that while the *tetQ* gene (1,859 base pairs) is present in the live organism, it is not present in the autoclaved *P. goldsteinii* RV-01 ingredient. The results of this analysis are presented in Figure 4, with lane 1 representing the live strain and lanes 2 – 6 representing various concentrations of autoclaved *P. goldsteinii* RV-01. Briefly, DNA was extracted from samples containing different amounts (1×10^6 - 1×10^{10} TFU) of autoclaved *P. goldsteinii* RV-01, followed by PCR and agarose gel electrophoresis analysis. The total DNA prepared from live *P. goldsteinii* RV-01 was used as the template for positive PCR control (lane 1), and deionized water was used as the negative PCR control (lane 7). The size of the *tetQ* PCR product is 1,859 base pairs and was not detected at any of the tested concentrations of autoclaved *P. goldsteinii* RV-01. Also, in the left image of Figure 4, there appears to be a signal near the top of lane 2. It was speculated that the lane 2 signal may have resulted from contamination of protein associated with the template DNA. Further purification of template DNA using the phenol/chloroform and ethanol precipitation method described by (Wright et al., 2017) removed this signal (right image of Figure 4). Therefore, the non-specific band identified in lane 2 was likely related

¹⁸ Accessed on April 22, 2025: [Homepage | Food Standards Australia New Zealand](#)

to protein contamination during the DNA template preparation and thus, this analysis confirms the absence of the *tetQ* gene in autoclaved *P. goldsteinii* RV-01.

To supplement the analysis presented above and in Figure 4, different amounts of *tetQ* DNA (10 pg to 0.0001 pg) and DNA from 0.001 g samples from three lots of autoclaved *P. goldsteinii* RV-01 were subjected to *tetQ* PCR (Figure 5.A). The intensity of the PCR product was quantified by densitometry. A linear correlation between the log amount of *tetQ* DNA (from 1 pg to 0.001 pg) and intensity of PCR product has been demonstrated (Figure 5.B). The standard curve shows that the *tetQ* gene is detectable at amounts above 0.001 pg. However, the *tetQ* gene was not detected in the DNA from 0.001 g samples of three lots of autoclaved *P. goldsteinii* RV-01. Therefore, the *tetQ* gene was not detected in the autoclaved *P. goldsteinii* RV-01 ingredient with a limit of detection (LOD) of 1 pg/g. It is therefore assumed that the autoclave and subsequent storage conditions for autoclaved *P. goldsteinii* RV-01 result in loss of integrity of the *tetQ* gene.

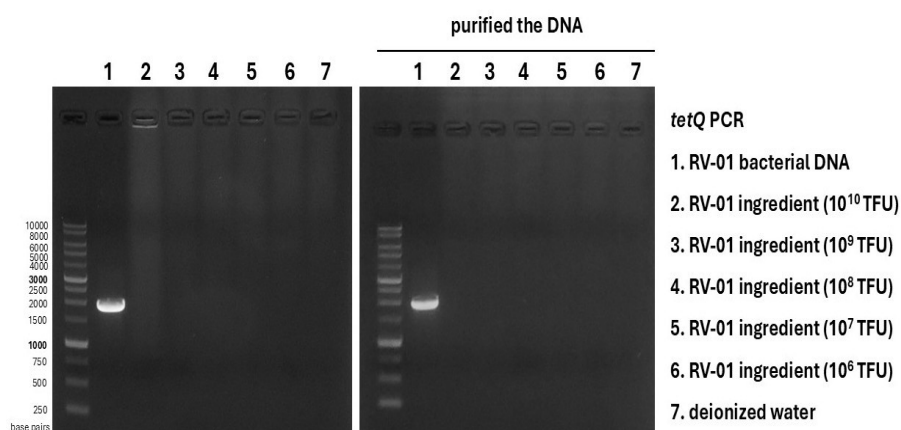


Figure 4. Gel Electrophoresis of the *tetQ* Gene in Autoclaved *P. goldsteinii* RV-01

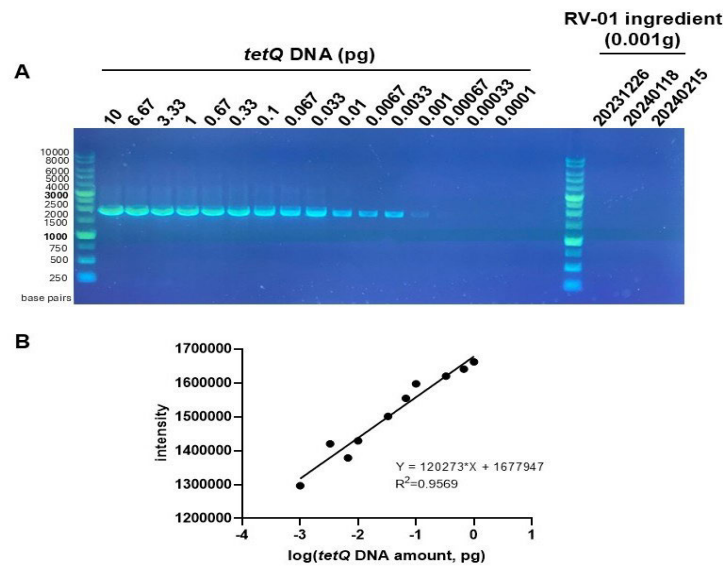


Figure 5. LOD Determination for the tetQ Gene and Analysis of Autoclaved *P. goldsteinii* RV-01

To further investigate the potential for antimicrobial resistance by *P. goldsteinii* RV-01, a minimum inhibitory concentration (MIC) test was performed, as presented in Table 12. The MIC values are compared to cut-off values for *Parabacteroides* species, established by the European Committee on Antimicrobial Susceptibility Testing (EUCAST).¹⁹ Further, a set of breakpoints for anaerobic bacteria were used from the Clinical and Laboratory Standards Institute (CLSI, 2020). As suggested by both organizations, *B. fragilis* ATCC 25285 was used as an anaerobic positive control strain. The results indicate that *P. goldsteinii* RV-01 is resistant to aminoglycosides (gentamycin, kanamycin, streptomycin, and apramycin), vancomycin, nalidixic acid, sulfonamide and trimethoprim. The MIC values for these antimicrobials were similar among the three *P. goldsteinii* strains and two other *Parabacteroides* species tested. Thus, resistance to these antimicrobials may be considered intrinsic, given the lack of genes which confer resistance to these antimicrobials in the genome of *P. goldsteinii* RV-01. In addition, *P. goldsteinii* RV-01 was susceptible to clindamycin, tetracycline, and chloramphenicol according to the appropriate breakpoints. The susceptibility of *P. goldsteinii* RV-01 to tetracycline indicates that the *tetQ* gene may have been mischaracterized or is inadequately expressed by the live organism.

Table 12. MIC Test with *P. goldsteinii* RV-01 and Related Organisms

Antibiotic	<i>B. fragilis</i> ATCC 25285	<i>P.</i> <i>goldsteinii</i> RV-01	<i>P.</i> <i>goldsteinii</i> MTS01	<i>P.</i> <i>goldsteinii</i> JCM13446	<i>P.</i> <i>distasonis</i> ATCC8503	<i>P. merdae</i> ATCC43184	CLSI Breakpoints for Anaerobic Bacteria	EUCAST Breakpoints for <i>Parabacteroides</i> Species ¹
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¹⁹ Accessed on April 22, 2025, from: [eucast: Clinical breakpoints and dosing of antibiotics](#)

MIC Values (µg/mL)								
Ampicillin	128	4	16	16	2	4	≤ 0.5	N/A
Vancomycin	32	16	>256	32	>256	32	N/A	N/A
Gentamycin	>256	>256	>256	>256	>256	>256	N/A	N/A
Kanamycin	>256	>256	>256	>256	>256	>256	N/A	N/A
Streptomycin	>256	>256	>256	>256	>256	>256	N/A	N/A
Erythromycin	8	8	8	8	1	8	N/A	N/A
Clindamycin	0.5	2	4	0.25	<0.125	0.25	≤ 2	≤ 4
Tetracycline	0.25	4	0.5	0.25	0.25	8	≤ 4	N/A
Chloramphenicol	8	4	8	8	4	4	≤ 8	N/A
Tylosine	2	0.5	1	0.5	0.5	0.5	N/A	N/A
Apramycin	>256	>256	>256	>256	>256	>256	N/A	N/A
Nalidixic acid	256	128	128	128	128	128	N/A	N/A
Sulfonamide	>256	>256	>256	>256	>256	>256	N/A	N/A
Trimethoprim	32	>256	>256	>256	128	>256	N/A	N/A

¹The EUCAST breakpoint for metronidazole is 4 µg/mL, but this was not tested against the organisms in this table and no other drug of this class (nitroimidazole) was evaluated. The EUCAST list does not include individual antimicrobials of the same classes as those tested in this table (except for clindamycin).

CLSI – Clinical and Laboratory Standards Institute; EUCAST – European Committee on Antimicrobial Susceptibility Testing; MIC – Minimum inhibitory concentration; mL – Milliliters; N/A – Not available; µg – Micrograms

b. Virulence Factors

Analysis of the *P. goldsteinii* RV-01 genome was conducted to screen for potential virulence factor genes using the Virulence Factor Database (VFDB). To identify potential virulence factor genes, thresholds of > 60% coverage and > 80% identity for virulence factor identity were applied based on relevant EFSA guidance (EFSA, 2021). There were no genes with > 60% coverage or > 80% identity to known virulence factors identified in the *P. goldsteinii* RV-01 genome.

c. Secondary Metabolites

The *P. goldsteinii* RV-01 genome was submitted to the antibiotics and secondary metabolite analysis shell (antiSMASH) tool version 7.0, as described by Blin et al. (2023), to identify any biosynthetic gene clusters for secondary metabolites. Secondary metabolites include compounds which confer a selective growth advantage to the organism. No biosynthetic gene clusters for secondary metabolites were identified in the *P. goldsteinii* RV-01 genome.

d. Biogenic Amines

The genome of *P. goldsteinii* RV-01 was submitted to the BLASTx search tool to identify the biogenic amine-producing genes histidine decarboxylase (*hdcA*), tyrosine decarboxylase (*tyrDC*), ornithine decarboxylase (*odc*), agmatine deiminase (*aguA* and *aguD*), and lysine decarboxylase (*ldc*). The potential for formation of biogenic amines is an important safety consideration because biogenic amines are associated with the toxicological characteristics of food poisoning in humans (JECFA,

2002). The genome of *P. goldsteinii* RV-01 contains one gene which encodes an agmatine deiminase homolog, that may catalyze the conversion of agmatine into N-carbamoylputrescine. However, in order to generate putrescine from N-carbamoylputrescine, another enzyme (N-carbamoylputrescine amidohydrolase) is required (Shimokawa et al., 2023). The activity of N-carbamoylputrescine amidohydrolase has been evaluated under various conditions and the results indicate that the activity is nullified at temperatures exceeding 70°C (Shimokawa et al., 2023). Given that the autoclave process for *P. goldsteinii* RV-01 involves heat treatment at 121°C, the autoclaved *P. goldsteinii* RV-01 ingredient is not expected to produce appreciable amounts of putrescine.

e. Mobile Genetic Elements

Considering the evidence of DNA degradation in the autoclaved *P. goldsteinii* RV-01 ingredient (Figure 4 and Figure 5), the presence of mobile genetic elements is not expected to pose a significant safety concern.

f. Allergenicity

A genomic analysis was conducted by ReviveBio, Co. to evaluate the allergenic potential of proteins present in *P. goldsteinii* RV-01. The GenBank *P. goldsteinii* RV-01 genome assembly, containing 5,241 protein coding sequences (CDS), was analyzed for potential allergenic proteins using Allermatch.²⁰ As per guidance from EFSA (2021), a threshold of 35% identity was used. Analyses were performed using full-length sequences, 80-amino acid alignment, and 6-amino acid exact matching. This analysis has been published by Lin et al. (2024).

In the full-length analysis, 33 hits with > 35% identity were identified. In the 80-amino acid alignment analysis, 174 hits were identified with > 35% identity. Finally, 13 hits were identified with a minimum of 6 exact amino acid matches. Of those 13 hits, 11 proteins were identified in the full-length sequence homology search and 80-amino acid alignment. Upon closer evaluation, three proteins (enolase²¹, adenosylhomocysteinase, and glyceraldehyde-3-phosphate dehydrogenase A) had > 50% identity in both full-length homology search and 80-amino acid alignment. However, an important consideration in the immunogenicity and cross-reactivity assessment of a protein is its structure. To this end, the authors concluded, “therefore, although sequence homology to known allergens was indicated, the autoclaved *P. goldsteinii* RV-01 ingredient is not expected to possess allergenic potential because the ingredient is prepared as an autoclaved and lyophilized powder which results in protein denaturation” (Lin et al., 2024).

However, contrary to the Lin et al. (2024) statement, enolase has been found to be resistant to heat, as it has been reported that enolase is found in cooked chicken meat samples (Wanniang et al., 2022). However, the enolase homolog in *P. goldsteinii* RV-01 is related to an inhalation allergen found in the fungus, *Rhodotorula mucilaginosa*.²² The activity and structure of enolase produced by

²⁰ Accessed on April 22, 2025, from: <http://allermatch.org>

²¹ Reported as “endolase” by Lin et al. (2024), but this has been confirmed to be in error, “enolase” is the correct name of the protein.

²² Accessed on April 22, 2025, from: [Rho m 1 Allergen Details](#)

the fungus and other microorganisms sharply declines and unfolds at temperatures exceeding 80°C (López-López et al., 2018; Zadvornyy et al., 2015). This temperature is below the 121°C used for the autoclaved *P. goldsteinii* RV-01 preparation. Therefore, the enolase homolog produced by *P. goldsteinii* RV-01 is not expected to pose a significant risk of allergenicity when consumed orally by humans ingesting the autoclaved *P. goldsteinii* RV-01 ingredient.

According to the World Health Organization and International Union of Immunological Societies (WHO/IUIS) Allergen Nomenclature Sub-Committee, adenosylhomocysteinase is a pollen allergen to which humans are exposed via inhalation.²³ The adenosylhomocysteinase homolog produced by *P. goldsteinii* RV-01 is related to adenosylhomocysteinase produced by the London Plane tree (*Platanus x hispanica*) which is also an inhalation allergen. A literature search conducted using PubMed on November 29, 2024, with the term, [Adenosylhomocysteinase and (allergy or allergen or allergenicity)] as well as several internet searches did not return any references which describe the allergenicity of adenosylhomocysteinase following oral exposure. Further, adenosylhomocysteinase is, "one of the most conserved enzymes in living organisms, including bacteria, nematodes, yeast, plants, insects, and vertebrates" (Vizán et al., 2021). Therefore, adenosylhomocysteinase produced by *P. goldsteinii* RV-01 is not expected to pose a significant risk of allergenicity.

Chen et al. (2022) reported that glyceraldehyde-3-phosphate dehydrogenase (GAPDH) is expressed by many bacteria, including the commensal organisms *Lactobacillus* (L.) *gasseri*, *L. reuteri*, *L. plantarum*, and *Bifidobacterium longum*, as well as the pathogens *Clostridium* (C.) *perfringens*, *C. difficile*, *Salmonella enterica*, and *E. coli*. Reyes et al. (2021) evaluated the potential allergenicity of a leghemoglobin preparation produced by *Pichia* (P.) *pastoris* which contains GAPDH and noted that, "GAPDH is a widely spread conserved protein whose homolog can be also found in many yeasts including *A. oryzae* and *P. roqueforti*, both which are widely used in the food industry, and from humans". Further, the authors demonstrated that the GAPDH protein was completely digested in an *in vitro* pepsin digestibility assay within 2 minutes. Ultimately, the authors concluded that "GAPDH from *P. pastoris* is unlikely to result in a food allergenic reaction". The GAPDH homolog produced by *P. goldsteinii* RV-01 is related to GAPDH produced by striped catfish (*Pangasianodon hypophthalmus*). However, Reuthers et al. (2020) have reported that unlike the native protein, the heated protein (95-100°C for 20 minutes) does not bind to IgE. Given that the autoclaved *P. goldsteinii* RV-01 ingredient is also subjected to heat treatment (121°C for 15 minutes), the GAPDH homolog is not expected to pose a significant risk of allergenicity.

As discussed in Part 6.A, *P. goldsteinii* produces hexosaminidase, a conserved enzyme identified in the Bacteroidota phylum, which possesses immunomodulatory properties. Searching the whole genome of *P. goldsteinii* RV-01 revealed a gene which is 97.3% identical with hexosaminidase. However, based on the results of an Allermatch search as described earlier in this section, the

²³ Accessed on April 22, 2025, from: [Pla a 5 Allergen Details](#)

hexosaminidase homolog in the *P. goldsteinii* RV-01 genome did not reveal any sequence homology to known allergens.

g. Cytotoxicity

To evaluate the potential cytotoxicity of *P. goldsteinii* RV-01, several doses of the live strain were administered to Human Embryonic Kidney 293 (HEK293) and Caco-2 cells with a multiplicity of infection (MOI) of 1, 10, 50, 100, 500, 1,000 or 2,000.²⁴ Cytotoxicity was measured using a lactate dehydrogenase (LDH) assay kit supplied by Promega, USA. No cytotoxicity was observed in either cell type at any of the doses tested. Results of this analysis are provided in Figure 6.

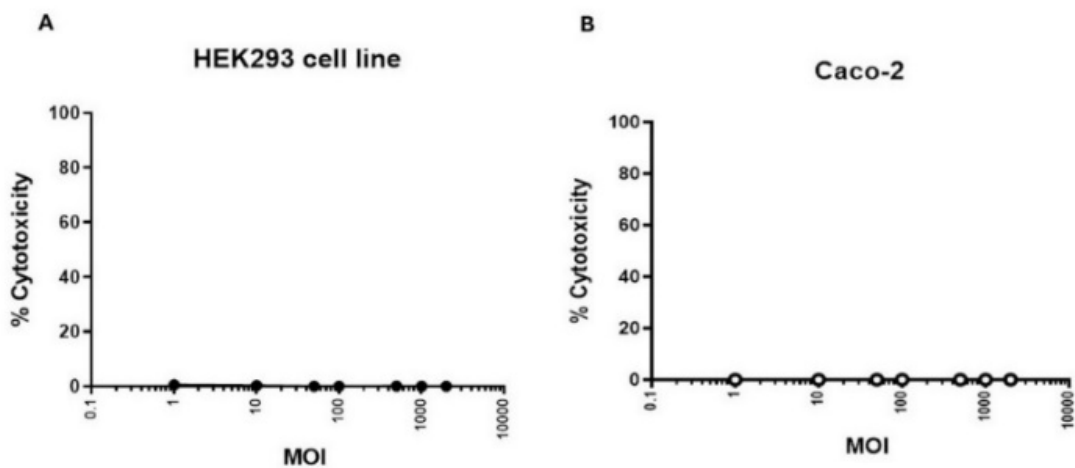


Figure 6. Cytotoxicity of Live *P. goldsteinii* RV-01

2. Genotoxicity Studies

The genotoxicity of autoclaved *P. goldsteinii* RV-01 has been evaluated in OECD Guideline *in vitro* reverse bacterial mutagenicity (Ames), *in vitro* chromosomal aberration, and *in vivo* micronucleus tests, as outlined in Table 13. The results of all tests were negative, indicating that autoclaved *P. goldsteinii* RV-01 does not pose a risk of genotoxicity.

Table 13. Genotoxicity Studies with Autoclaved *P. goldsteinii* RV-01

²⁴ MOI is the ratio of test cells to target cells. For example, an MOI of 10 means there are 10-times more *P. goldsteinii* RV-01 cells than HEK293 or Caco-2 cells.

Study Design	Test System	Doses	Results	Reference
Bacterial reverse mutation (Ames) test according to GLP and OECD 471	<i>Salmonella typhimurium</i> TA97a, TA98, TA100, TA102, and TA1535 with or without metabolic activation	0.3125, 0.625, 1.25, 2.5, or 5 mg/plate	There were no significant changes in the number of revertant colonies between the control plates and any concentration of the test material.	Lin et al. (2024)
<i>In vitro</i> chromosomal aberration test according to GLP and OECD 473	Chinese hamster ovary cells with or without metabolic activation	0.3125, 0.625, 1.25, 2.5, or 5 mg/mL	There were no significant changes in the percentage of cells with chromosomal aberrations between the control and any concentration of the test material.	Lin et al. (2024)
<i>In vivo</i> micronucleus test according to GLP and OECD 474	Male ICR mice	500, 1,000 or 2,000 mg/kg bw	There were no significant changes in the number of micronucleated reticulocytes or the frequency (%) of micronucleated reticulocytes between treatment and control groups.	Lin et al. (2024)

bw – Body weight; GLP – Good Laboratory Practice; kg – Kilograms; mg – Milligrams; mL – Milliliters; OECD – Organization for Economic Cooperation and Development

3. Subacute and Subchronic Toxicity Studies

Lin et al. (2024) conducted a 28-day repeat dose toxicity study with autoclaved *P. goldsteinii* RV-01 in male and female Sprague Dawley rats, in compliance with GLP and according to OECD Guideline 407. The animals (n = 5/sex/group) received vehicle control or autoclaved *P. goldsteinii* RV-01 at 74, 370, or 1,500 mg/kg bw/day, corresponding to intakes of 4×10^9 , 2×10^{10} , or 8.1×10^{10} TFU/kg bw/day, respectively, for 28 consecutive days via oral gavage. No mortality occurred and no clinically adverse observations were made during the study. There were no statistically significant changes in body weight, food intake, water intake, organ weights, or organ to brain weight ratios between treatment and control groups. No gross lesions were observed in any of the animals and no treatment-related histopathological findings were observed in high dose animals. Therefore, the no observed adverse effect level (NOAEL) of autoclaved *P. goldsteinii* RV-01 was determined to be 1,500 mg/kg bw/day, or 8.1×10^{10} TFU/kg bw/day, the highest dose level tested.

GA also reviewed the full study report²⁵, and the following additional information was identified.

- Control and high dose recovery groups (n = 5/sex/group) were observed for 14 days following dosing to evaluate the reversibility or delayed onset of potentially adverse effects.

²⁵ Accessed on April 22, 2025, from: [28-Day Oral Toxicity – 微大科技股份有限公司](#)

- No changes in ophthalmological observations were observed between the treatment and control groups.
- Males in the middle dose group had a statistically significant (2%) decrease in mean corpuscular hemoglobin concentration (MCHC) compared to controls, but the mean MCHC concentration in the middle dose group (33.66 ± 0.29 g/dL) did not deviate from the reference range of 33.0 - 38.0 g/dL (Derelanko, 2000).
- No other changes in hematological, coagulation, or clinical chemistry parameters were observed between treatment and control groups.
- Renal tubular mineralization was observed in 2/5 females in the control group, 4/5 females in the high dose group, 4/5 females in the recovery control group, and 5/5 females in the high dose recovery group. This finding was not observed in any of the control or high dose males in the main study or recovery groups. This observation in females was not determined to be treatment-related by the authors of the study report.²⁶ Rao (2002) reports that renal tubular mineralization is more common in female Sprague-Dawley, Wistar, and Fisher 344 rats compared to males of the same strains. This finding has been observed in up to 100% of female Wistar rats (6/6) and may be dependent on the calcium:phosphorus ratio of the diet (Rao, 2002). NRC (1995) also reports that female Sprague-Dawley and Wistar rats are particularly susceptible to this type of lesion and that a calcium:phosphorus molar ratio ≥ 1.3 in the diet is “needed to prevent nephrocalcinosis in female rats.” Other factors which may contribute to an increased risk of renal mineralization in female rats are the source of protein and carbohydrate. For example, the substitution of lactalbumin with casein while maintaining the amount of phosphorus in the diet was associated with decreased calcium accumulation in the kidneys of Sprague-Dawley rats (NRC, 1995). Further, dietary fructose, as opposed to glucose, stimulates nephrocalcinosis in female rats by increasing concentrations of urinary phosphorus and magnesium and lowering the urine pH (NRC, 1995). However, it is unclear if any of these conditions apply to the study conducted by Lin et al. (2024) because information about the composition of the diet was not supplied in the available documentation.

None of these observations change the conclusion that the NOAEL of autoclaved *P. goldsteinii* RV-01 is 1,500 mg/kg bw/day (8.1×10^{10} TFU/kg bw/day) in male and female rats.

Lin et al. (2024) also conducted a 90-day repeat dose toxicity study with autoclaved *P. goldsteinii* RV-01 in compliance with GLP and according to OECD Guideline 408. This 90-day study in rats represents pivotal evidence of safety for autoclaved *P. goldsteinii* RV-01. Male and female Sprague-Dawley rats ($n = 10/\text{sex}/\text{group}$) received vehicle control or autoclaved *P. goldsteinii* RV-01 at doses of 74, 370, or 1,500 mg/kg bw/day, corresponding to 4×10^9 , 2×10^{10} , or 8.1×10^{10} TFU/kg bw/day, via oral gavage. The authors stated that there were no significant clinically adverse observations,

²⁶ Accessed on April 22, 2025, from: [28-Day Oral Toxicity – 微大科技股份有限公司](#)

ophthalmological observations, or changes in functional observational battery and “no positive correlations in the degree and incidence rate of the histological lesions” between treatment and control groups. The NOAEL of autoclaved *P. goldsteinii* RV-01 was determined to be 1,500 mg/kg bw/day, or 8.1×10^{10} TFU/kg bw/day in male and female rats, the highest dose level tested.

GA also reviewed the full study report²⁷, and the following additional information was identified.

- Control and high dose recovery groups (n = 5/sex/group) were observed for 14 days following dosing to evaluate the reversibility or delayed onset of potentially adverse effects.
- No mortality occurred in any of the groups.
- No differences in bodyweight between treatment and control groups during the main study period. During the recovery period, females in the high dose group had a significant increase in bodyweight compared to recovery controls, but this change was not observed at any point during the main study period and was thus determined to not be of toxicological significance. Further, there were no changes in food intake between treatment and control groups throughout the main study or recovery period.
- No changes in organ weights or relative organ weights (relative to brain weight) in any of the treatment groups compared to the controls, except for a decrease in thyroid weight (both absolute and relative) in the low dose males compared to controls. However, this finding was not dose-dependent and there were no accompanying pathological findings, and therefore this was determined not to be toxicologically significant.
- No statistically significant differences in serum thyroid stimulating hormone (TSH), triiodothyronine (T3), or thyroxine (T4) between any of the treatment groups and controls in the main study or recovery period.
- No changes in estrus cycle parameters in any of the treated females compared to controls.
- None of the urinalysis parameters revealed any changes between the treatment and control groups.
- Observed changes in hematology parameters are as follows:
 - Low and middle dose males had decreased MCHC compared to controls. However, this change was not observed in the high dose group and the mean values in the low dose (34.38 ± 0.39 g/dL) and middle dose (34.32 ± 0.45 g/dL) groups did not deviate from the reference range of 33.0 - 38.0 g/dL (Derelanko, 2000).

²⁷ Accessed on April 22, 2025, from: [90-Day Oral Toxicity – 微大科技股份有限公司](#)

- High dose recovery females had a 3% increase in mean corpuscular volume (MCV) compared to controls, but the values in the recovery (53.76 ± 0.89 fL) and control (51.90 ± 1.20 fL) groups were at the low end or below the reference range of 53.0 - 63.0 fL, respectively (Derelanko, 2000).
 - Low dose females had increased hemoglobin compared to controls, but this was not dose-dependent, and the mean value in the low dose group (15.55 ± 0.44 g/dL) did not deviate from the reference range of 14.0 - 17.0 g/dL (Derelanko, 2000).
- High dose males had a 14% decrease in prothrombin time (PT) compared to controls in the main study. However, this change was not observed in the high dose recovery males compared to their respective controls. Further, the mean PT value in the high dose males (12.08 ± 0.47 seconds) did not deviate from the reference range of 9.0 – 14.0 seconds (Derelanko, 2000). There was no effect of any dose of the test material on activated partial thromboplastin time (APTT) in males.
- There was a statistically significant increase in APTT in middle dose females compared to controls. This finding was not dose-dependent and was not observed in the high dose recovery females compared to their respective controls. The mean APTT value in the middle dose females (20.17 seconds) did deviate from the reference range of 13.0 – 18.0 seconds (Derelanko, 2000), but each of the control and treatment groups in the main study and recovery groups exceeded the reference range. This suggests that the animals used in the study were predisposed to having slightly higher APTT than rats of the same age and strain by some factor unrelated to treatment.
- Reduced serum aspartate aminotransferase (AST) was observed in middle dose males compared to controls, but this change was not observed in the main or recovery high dose groups and the mean value in the middle dose group (62.2 ± 6.5 U/L) did not deviate from the reference range of 45.0 - 90.0 U/L (Derelanko, 2000).
- The incidences of renal tubular mineralization in control and high dose treated females were 6/10 and 4/10, respectively, and the incidence was 2/5 in both the recovery control and recovery high dose females, indicating a lack of treatment-related effect. The incidence of renal tubular mineralization in males was 1/10 and 0/10 in the control and high dose main study groups, respectively, and 1/5 and 0/5 in the recovery control and recovery high dose groups, respectively. These results support the findings of Rao (2002) and NRC (1995) that control female Sprague Dawley rats spontaneously develop renal tubular mineralization and show that treatment with autoclaved *P. goldsteinii* RV-01 does not promote this lesion.

None of these observations change the conclusion that the NOAEL of autoclaved *P. goldsteinii* RV-01 is 1,500 mg/kg bw/day (8.1×10^{10} TFU/kg bw/day) in male and female rats.

4. Other Animal Safety Studies

Lai et al. (2022) administered live *P. goldsteinii* MTS01 to male Balb/c mice (n = 10/group) for 12 weeks. The dose of *P. goldsteinii* MTS01 was 2×10^8 CFU/day for the first 10 weeks, followed by 1×10^9 CFU/day for the next 2 weeks, via oral gavage. Bodyweight and body temperature did not significantly differ between treatment and control groups. Serum triglycerides (TG), total cholesterol (TC), low density lipoprotein (LDL), and very low-density lipoprotein (VLDL) did not significantly differ between groups. The *P. goldsteinii* MTS01 group had reduced serum high density lipoprotein (HDL) compared to controls. Histopathological analysis of gastric tissue revealed no differences between the *P. goldsteinii* MTS01 and control groups. Given that the test material was live, and a different strain of *P. goldsteinii*, this effect on serum HDL is not expected to occur following oral exposure to autoclaved *P. goldsteinii* RV-01. Further, there was no change in serum HDL between treatment and control groups in the 90-day toxicity study of autoclaved *P. goldsteinii* RV-01. Therefore, this study provides supportive evidence of safety related to the intended use of autoclaved *P. goldsteinii* RV-01.

5. Human Safety Studies

No clinical studies with *P. goldsteinii*, including live or autoclaved *P. goldsteinii* RV-01, have been identified in the published literature.

Case reports involving infection with live *P. goldsteinii* have been reported in the published literature. For example, an 82-year-old woman with a history of diverticular disease became infected with several bacteria, including *P. goldsteinii*, after colonic perforation (Awadel-Kariem et al., 2010). This patient died following 30 days of treatment for the infection which included several pathogenic bacteria (including streptococci and *Candida*) as well as blood pressure abnormalities. Another patient was likely immunocompromised due to treatment for lymphoma when they became infected with several bacteria, including *P. goldsteinii* (Cobo et al., 2022). The patient was treated with piperacillin-tazobactam (4 g/8 hours IV) for 10 days and fully recovered from the infection. Finally, another patient became infected with *P. goldsteinii* following perineal lacerations, cervical lacerations, and postpartum hemorrhage during childbirth (Peng et al., 2024). This patient underwent uterine curettage and treatment via uterine cavity irrigation with metronidazole and intramuscular injection of gentamicin and dexamethasone. The patient recovered and was discharged 30 days after admission. None of the reported cases involved oral intake of *P. goldsteinii* prior to the reported infections.

Since the ingredient contains autoclaved *P. goldsteinii* RV-01 and the specification for the live strain is < 10 CFU/g, the ingredient is not expected to pose a significant risk of infection.

F. GRAS Criteria

FDA defines “safe” or “safety” as it applies to food ingredients as:

“...reasonable certainty in the minds of competent scientists that the substance is not harmful under the intended conditions of use.”²⁸

Amplification is provided in that the conclusion of safety is to include probable consumption of the substance in question, the cumulative effect of the substance and appropriate safety factors. It is FDA’s operational definition of safety that serves as the framework against which this evaluation is provided.

Furthermore, in discussing GRAS criteria, FDA notes that:

“...General recognition of safety requires common knowledge, throughout the expert scientific community knowledgeable about the safety of substances directly or indirectly added to food, that there is reasonable certainty that the substance is not harmful under the conditions of its intended use.”

“‘Common knowledge’ can be based on either “scientific procedures” or on experience based on common use of a substance in food prior to January 1, 1958.”
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FDA discusses in more detail what is meant by the requirement of general knowledge and acceptance of pertinent information within the scientific community, i.e., the so-called “common knowledge element,” in terms of the two following component elements:

- Data and information relied upon to establish safety must be generally available, and this is most commonly established by utilizing published, peer-reviewed scientific journals; and
- There must be a basis to conclude that there is consensus (but not unanimity) among qualified scientists about the safety of the substance for its intended use, and this is established by relying upon secondary scientific literature such as published review articles, textbooks, or compendia, or by obtaining opinions of expert panels or opinions from authoritative bodies, such as JECFA and the National Academy of Sciences.

General recognition of safety based upon scientific procedures shall require the same quantity and quality of scientific evidence as is required to obtain approval of a food additive. General recognition of safety through scientific procedures shall be based upon the application of generally available and accepted scientific data, information, or methods, which ordinarily are published, as well as the application of scientific principles, and may be corroborated by the application of unpublished scientific data, information, or methods.

²⁸ See 21 CFR 170.3 (e)(i) and 81 FR 54959 Available at: <https://www.federalregister.gov/documents/2016/08/17/2016-19164/substances-generally-recognized-as-safe>

The apparent imprecision of the terms “appreciable,” “at the time,” and “reasonable certainty” demonstrates that the FDA recognizes the impossibility of providing absolute safety in this or any other area (Lu, 1988; Renwick, 1990; Rulis and Levitt, 2009).

As noted below, this safety assessment to ascertain GRAS status for the specified food uses meets FDA criteria for reasonable certainty of no harm by considering both the technical and common knowledge elements.

G. Expert Panel Findings on Safety of ReviveBio, Co.’s Autoclaved *P. goldsteinii* RV-01

An evaluation of the safety and GRAS status of the intended use of ReviveBio, Co.’s autoclaved *P. goldsteinii* RV-01 has been conducted by an Expert Panel convened by GRAS Associates; the Panel consisted of Gregor Reid, PhD; Daniel O’Sullivan, PhD; and Kayla Preece, ND, DABT as Panel Chair. The Expert Panel reviewed ReviveBio, Co.’s dossier as well as other publicly available information. The individuals who served as Expert Panelists are qualified to evaluate the safety of food and food ingredients by merit of scientific training and experience. The GRAS Expert Panel report is provided in Appendix 7.

H. Common Knowledge Elements for GRAS Conclusions

The first common knowledge element for a GRAS conclusion requires that data and information relied upon to establish safety must be generally available; this is commonly established by utilizing studies published in peer-reviewed scientific journals. The second common knowledge element for a GRAS conclusion requires that consensus exists within the broader scientific community.

1. Public Availability of Scientific Information

The regulatory framework for determining whether a substance is GRAS is in 21 CFR 170.30, which states that GRAS status through scientific procedures shall ordinarily be based upon published studies, which may be corroborated by unpublished studies and other data and information. These criteria have been applied to the existing data for autoclaved *P. goldsteinii* RV-01.

All studies which have been reviewed and referenced in this dossier are published in the scientific literature or are available online, including all pivotal safety information summarized throughout Part 6.

2. Scientific Consensus

The second common knowledge element for a GRAS conclusion requires that there be a basis to conclude that consensus exists among qualified scientists about the safety of the substance for its intended use.

ReviveBio, Co.’s autoclaved *P. goldsteinii* RV-01 ingredient is manufactured according to Taiwan’s cGMP, ISO 22000:2018, and HACCP principles. The specifications and analysis demonstrate

sufficient quality control of the ingredient, such that potential contaminants are not expected to be present at levels which would pose a significant risk of toxicity. In addition, autoclaved *P. goldsteinii* RV-01 has been analyzed for SCFAs given the inherent properties of the live organism and presence of carbohydrates in the final ingredient. The autoclaved *P. goldsteinii* RV-01 ingredient contains primarily acetic and propionic acids, as well as a smaller amount of formic acid, which are not expected to be present at levels which would adversely affect its safety. Further, the stability of autoclaved *P. goldsteinii* RV-01 has been demonstrated over 3 months under accelerated conditions and 6 months under ambient conditions. No significant history of use or regulatory history was identified for the species or strain. However, ReviveBio, Co. has conducted several *in silico*, *in vitro*, and *in vivo* studies to demonstrate the safety of its autoclaved *P. goldsteinii* RV-01 ingredient.

The potential for antimicrobial resistance was evaluated by analyzing the genome of *P. goldsteinii* RV-01 for genes which are known to confer resistance to antimicrobials. One gene associated with tetracycline resistance (*tetQ*) was identified in the genome; however, the gene is not detected in the autoclaved *P. goldsteinii* RV-01 ingredient. Further, in a MIC test with live *P. goldsteinii* RV-01, the organism was susceptible to tetracycline. Therefore, autoclaved *P. goldsteinii* RV-01 is not expected to demonstrate resistance to tetracycline or transfer any genes associated with antimicrobial resistance to other microorganisms in the human gut. In the same MIC test, *P. goldsteinii* RV-01 demonstrated resistance to several antimicrobials. However, this is considered intrinsic to the species as other strains of *P. goldsteinii* and other *Parabacteroides* species demonstrated the same resistance profile, and no known genetic determinants of resistance to these antimicrobials were identified in the genome of *P. goldsteinii* RV-01.

Analysis of the *P. goldsteinii* RV-01 genome using the VFDB and antiSMASH tools revealed no known virulence factors or biosynthetic gene clusters for secondary metabolites, respectively. Analysis of the genome for biogenic amine-producing genes revealed an agmatine deiminase homolog, which may catalyze the conversion of agmatine into N-carbamoylputrescine. However, since the ingredient is autoclaved at 121°C, it is not expected to maintain the enzymatic activity necessary to produce appreciable amounts of putrescine (the end product of metabolism of agmatine and N-carbamoylputrescine). Further, live *P. goldsteinii* RV-01 is not cytotoxic to HEK293 or Caco-2 cells at a MOI up to 2,000.

In addition, several nonclinical safety studies have been conducted with ReviveBio, Co.'s autoclaved *P. goldsteinii* RV-01 ingredient. Genotoxicity studies including Ames, *in vitro* chromosomal aberration, and *in vivo* micronucleus tests conducted in compliance with GLP and according to the appropriate OECD guidelines have consistently been negative, indicating a lack of genotoxic potential for the ingredient. Further, 28- and 90-day toxicity studies in compliance with GLP and according to the appropriate OECD guidelines support a NOAEL of 1,500 mg/kg bw/day (8.1×10^{10} TFU/kg bw/day) which was the highest dose level tested in male and female rats. The highest 90th percentile intake on a bodyweight basis is 19.8 mg/kg bw/day in male children aged 1-5 years. The 90th percentile intakes in adult males (population with the highest absolute intake) and the general population are 7.1 and 8.2 mg/kg bw/day, respectively. The absolute intakes in male children aged 1-5 years, adult males,

and the general population at the 90th percentile are 342.7, 618.5, and 475.9 mg/day, respectively. Based on the NOAEL of autoclaved *P. goldsteinii* RV-01 of 1,500 mg/kg bw/day, the margins of safety are 76 in male children aged 1-5 years, 211 in adult males, and 183 in the general population.

In summary, a compelling case can be made that scientific consensus exists regarding the safety of ReviveBio, Co.'s autoclaved *P. goldsteinii* RV-01 ingredient.

I. Discussion of Information Inconsistent with GRAS Conclusion

The authors of this GRAS conclusion are not aware of any information that would be inconsistent with a finding that the proposed use of autoclaved *P. goldsteinii* RV-01 in food is generally recognized as safe.

J. Conclusion

In consideration of the aggregate safety information available, ReviveBio, Co. concludes that autoclaved *P. goldsteinii* RV-01 as defined in the subject notification is expected to be safe for use as an ingredient in sports and energy drinks, ready-to-eat cereals (low sugar), yogurt, fermented milk products, ice cream, cereal bars, nutrition bars, nut spreads, and candy containing chocolate at levels up to 1×10^{10} TFU/serving, or 185 mg/serving. The intended use does not include infant formula or USDA-regulated food. The mean and 90th percentile intakes of autoclaved *P. goldsteinii* RV-01 in the general population are 223.8 mg/day (1.0×10^{10} TFU/day) and 475.9 mg/day (2.2×10^{10} TFU/day), respectively.

The weight of the publicly available evidence from studies with autoclaved *P. goldsteinii* RV-01 provide a basis upon which to conclude that the proposed uses of autoclaved *P. goldsteinii* RV-01 as described in this dossier satisfy the safety standard of Reasonable Certainty of No Harm and are safe. Based on the pivotal, published data and information that are generally available, one may conclude that the proposed uses of autoclaved *P. goldsteinii* RV-01 as produced consistent with ISO 22000:2018 and meeting the food grade specifications presented above, are Generally Recognized As Safe (GRAS) based on scientific procedures. Support for these conclusions by a consensus of qualified experts in the general scientific community is provided in Appendix 7 (Expert Panel Report).

Accordingly, autoclaved *P. goldsteinii* RV-01 as produced in accordance with ISO 22000:2018 and when it meets those specifications declared within the subject notification meets FDA's definition of safety in that there is "reasonable certainty of no harm under the intended conditions of use" as described herein and, therefore, is Generally Recognized as Safe (GRAS).

PART 7. LIST OF SUPPORTING DATA AND INFORMATION IN THE GRAS NOTICE.

A. References

1. List of Acronyms

ADI	Average daily intake
ADME	Absorption, Distribution, Metabolism and Excretion
ANI	Average nucleotide identity
antiSMASH	Antibiotics and secondary metabolite analysis shell
APTT	Activated partial thromboplastin time
AST	Aspartate aminotransferase
BLAST	Basic Local Alignment Search Tool
bw	Body weight
C	Celsius
CARD	Comprehensive Antibiotic Resistance Database
CDC	Center for Disease Control and Prevention
CDS	Protein coding sequences
CFU	Colony forming units
cGMP	Current Good Manufacturing Practices
dL	Deciliters
DSLID	Dietary Supplement and Labeling Database
EUCAST	European Committee on Antimicrobial Susceptibility Testing
FALCPA	The Food Allergen Labeling and Consumer Protection Act of 2004
FASTER	Food Allergy Safety, Treatment, Education and Research Act of 2021
FD&C Act	Federal Food Drug and Cosmetics Act
FDA	U.S. Food and Drug Administration
FSANZ	Food Standards Australia and New Zealand
g	Grams
GA	GRAS Associates
GAPDH	Glyceraldehyde-3-Phosphate Dehydrogenase
GC	Guanine-cytosine
GRAS	Generally Recognized as Safe
HACCP	Hazard Analysis and Critical Control Point
HDL	High density lipoprotein
HEK293	Human Embryonic Kidney 293
JECFA	Joint FAO/WHO Expert Committee on Food Additives
kg	Kilograms
L	Liters
LC-MS	Liquid chromatography-mass spectrometry
LDH	Lactate dehydrogenase
LDL	Low density lipoprotein
MCHC	Mean corpuscular hemoglobin concentrations
MCV	Mean corpuscular volume
mg	Milligrams
MIC	Minimal inhibitory concentration
mL	Milliliters
mm	Millimeters
MOI	Multiplicity of infection
NDIN	New Dietary Ingredient Notification
NHANES	National Health and Nutrition Examination Survey

NOAEL	No observed adverse effect level
PCR	Polymerase chain reaction
pg	Picograms
ppm	Parts per million
PT	Prothrombin time
QC	Quality control
SCFAs	Short chain fatty acids
T3	Triiodothyronine
T4	Thyroxine
TC	Total cholesterol
TFU	Total fluorescent units
TG	Triglycerides
TSH	Thyroid stimulating hormone
UL	Tolerable upper limit
VFDB	Virulence Factor Database
VLDL	Very low-density lipoprotein
WCB	Working Cell Bank
WHO/IUIS	World Health Organization and International Union of Immunological Societies

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B. Appendices

Appendix 1 Supplier Statement



No. 81 Gaueyang N. Road, Longtan District, Taoyuan City, TAIWAN

Ceclaration

This is to confirm that Microbio, Co. Ltd. (LongTan GMP Factory) is the commissioned producer for production of REVIVEBIO.CO.' autoclaved *P. goldsteinii* RV-01 product. All procedures follow the regulations based on GMP quality standards.

Signed by Person in charge: _____

20250/10



Appendix 2 cGMP Certificate



衛生福利部
MINISTRY OF HEALTH AND WELFARE

CERTIFICATE OF GOOD MANUFACTURING PRACTICE

Issue Date: JUL 20 2023

Issued following an inspection in accordance with Article 57 of the Pharmaceutical Affairs Law and relevant Regulations of the Republic of China (Taiwan).

Importing Country (ies): United States of America, Canada, European Union.
The competent authority of the Republic of China confirms the following:

The manufacturer: MICROBIO CO., LTD. LONGTAN FACTORY
Site address: No.81, Gaoyang N. Road, Longtan District, Taoyuan City, Taiwan, R.O.C.
Manufacturer's licence number: (C)0000011

is the manufacturer of medicinal products for human use.

From the knowledge gained during inspection performed on Apr. 27, 2023
, it is considered that the manufacturer complies with the Pharmaceutical Good Manufacturing Practice Regulations for medicinal products.

This certificate is valid until Jun. 20, 2025.
This certificate may be revoked at anytime as warranted.

Signed by



Yi-Tsau Huang, M.D., Ph.D.
Director-General
Department of Chinese Medicine and Pharmacy

Under the delegated authority of
Jui-Yuan Hsueh, M.D., LL.M.
Minister
Ministry of Health and Welfare
Republic of China (Taiwan)

Appendix 3 ISO 22000:2018 Certificate

Certificate TW11/10537 / 證書 TW11/10537

The management system of / 下述組織

MICROBIO CO., LTD. LONGTAN FACTORY
中天生物科技股份有限公司龍潭廠

No. 81, GaoYang N. Road, Longtan District, Taoyuan City, Taiwan, R.O.C.
桃園市龍潭區高平里高楊北路 81 號

has been assessed and certified as meeting the requirements of
的管理系統已經過審核，並被證明符合下述要求
ISO 22000:2018

For the following activities / 所涵蓋的活動範圍

1. Manufacturing of ambient stable hermetically sealed bottled fermented concentrated drinks of dietary supplements
2. Manufacturing of ambient stable hermetically sealed herbal drinks of dietary supplements packed in bottles and sachets
3. Manufacturing of tablet, and capsule types of dietary supplements packed in bottles
4. Manufacturing of powder types of dietary supplements packed in bottles and sachets.

1. 製造常溫穩定密封瓶裝發酵濃縮膳食補充劑飲料。
2. 製造常溫穩定密封瓶裝及袋裝草本膳食補充劑飲料。
3. 製造瓶裝的片劑和膠囊劑膳食補充劑。
4. 製造瓶裝及袋裝的粉狀膳食補充劑。

Food Category / 食品類別-CIV - Processing of ambient stable products
This certificate is valid from 01 August 2023 until 01 August 2026 and remains valid subject to satisfactory surveillance audits.
Issue 5. Certified since 01 August 2011

此證書的有效期限自 2023 年 08 月 01 日至 2026 年 08 月 01 日且其有效性應繫於持續符合的定期稽核
版次 5. 原始註冊日期 2011 年 08 月 01 日

Authorised by / 簽署
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本文件為真實的電子證書，僅供客戶用於其商業用途。此電子證書允許其印刷版本均應視為副本。本文件由本公司依據可取之 Terms and Conditions | SGS 中之 SGS 驗證服務條款條款頒發。敬請注意其中包含的責任範圍、賠償和司法管轄事項。本文件受版權保護，任何未經授權而對此文件內容或外觀作出變更、偽造或篡改皆屬違法。

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Appendix 4 HACCP Certificate

Certificate TW11/10538 / 證書 TW11/10538

The management system of / 下述組織

MICROBIO CO., LTD. LONGTAN FACTORY
中天生物科技股份有限公司龍潭廠

No. 81, GaoYang N. Road, Longtan District, Taoyuan City, Taiwan, R.O.C.
桃園市龍潭區高平里高楊北路 81 號

has been assessed and certified as meeting the requirements of
的管理系統已經過審核，並被證明符合下述要求
**HACCP Certification (Hazard Analysis and Critical Control Point) Based upon Codex Alimentarius
General Principles of Food Hygiene CXC 1-1969 (2020)**

For the following activities / 所涵蓋的活動範圍

1. Manufacturing of ambient stable hermetically sealed bottled fermented concentrated drinks of dietary supplements
2. Manufacturing of ambient stable hermetically sealed herbal drinks of dietary supplements packed in bottles and sachets
3. Manufacturing of tablet, and capsule types of dietary supplements packed in bottles
4. Manufacturing of powder types of dietary supplements packed in bottles and sachets.

1. 製造常溫穩定密封瓶裝發酵濃縮膳食補充劑飲料。
2. 製造常溫穩定密封瓶裝及袋裝草本膳食補充劑飲料。
3. 製造瓶裝的片劑和膠囊劑膳食補充劑。
4. 製造瓶裝及袋裝的粉狀膳食補充劑。

Food Category / 食品類別-CIV - Processing of ambient stable products
This certificate is valid from 01 August 2023 until 01 August 2026 and remains valid subject to satisfactory surveillance audits.
Issue 6. Certified since 01 August 2011

此證書的有效期自 2023 年 08 月 01 日至 2026 年 08 月 01 日且其有效性應繫於持續符合的定期稽核
版次 6. 原始註冊日期 2011 年 08 月 01 日

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Appendix 5 Raw Materials Attestation



2024/12/26

14F-5, No. 3, Park Street, Nangang Dist., Taipei City 115, Taiwan (R.O.C)

To whom it may concern

This is to affirm that "All raw materials, processing aids, and food contact substances used to manufacture autoclaved *P. goldsteinii* RV-01 are food-grade ingredients permitted by U.S. regulations or have previously been determined to be GRAS for their respective uses".

Best Regards

Hsin-Chih Lai

General Manager

REVIVEBIO.CO.

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Appendix 6 Intake Analysis Report

Proposed Use and Food Codes Utilized for Intake Calculation

The dietary exposure distributions were calculated using the Creme Food Safety® modelⁱ, a scientific cloud-based software service designed and developed to calculate dietary intakes of foods, chemicals, and nutrients in populations of consumers. This is achieved by linking food consumption data from the What We Eat In America (WWEIA) portion of the National Health and Nutrition Examination Survey (NHANES) to the appropriate food composition and chemical occurrence data using a number of validated and published models, available upon request from Crème Global (<https://www.cremeglobal.com/>). Output calculations include daily average intakes, acute exposures, as well as population statistics such as mean, percentiles, standard errors, and confidence intervals. Results are output for “Consumers-Only” (i.e., consumers of the food / substance of interest), and “Total Population” (consumers and non-consumers). Results of the exposure assessment are given in absolute terms (mg per day) as well as relative to the consumer’s body weight (mg per kg body weight per day). The per unit of bodyweight exposure is calculated on a subject level using the bodyweight recorded by the NHANES data.

The proposed uses of autoclaved *P. goldsteinii* RV-01, are provided in Table 1 below. Autoclaved *P. goldsteinii* RV-01 is intended to be used as an ingredient in the following conventional foods: sports and energy drinks, ready-to-eat cereal (low sugar), yogurt, kefir, ice cream and frozen desserts, cereal and nutrition bars, nut spreads, and candy containing chocolate. The target amount is up to 1×10^{10} total fluorescent units (TFU) or approximately 185 mg per daily serving in the finished food product. The food codes utilized to calculate the estimated daily intake were selected from the NHANES 2017 – 2020 Pre-pandemic survey. Food codes that most appropriately match the intended use were selected. The final food code selections are found in Table 2.

Table 1. Decision Criteria for Food Codes

Aggregate Category	Maximum Proposed Use Per Serving (mg)	Maximum Proposed Concentration (mg/100 g)	Decision Criteria
Sports and Energy Drinks	185	51.4	Included all sports and energy drinks and diet sports and energy drinks food codes except fluid replacement codes and drinks intended for children such as Pedialyte.
Ready-to-eat Cereal (Low Sugar)	185	1233.3	Included all ready-to-eat cereal (low sugar) food codes.
Yogurt	185	108.8	Included all yogurt and Greek yogurt food codes except for those intended for children.
Milk, Low-fat	185	77.1	Included only fermented milk product food codes (kefir).

Aggregate Category	Maximum Proposed Use Per Serving (mg)	Maximum Proposed Concentration (mg/100 g)	Decision Criteria
Ice Cream and Frozen Desserts	185	115.6	Included all ice cream and frozen dessert food codes.
Cereal Bars	185	462.5	Included all cereal bar food codes.
Nutrition Bars	185	462.5	Included all nutrition bar food codes.
Nut and Seeds	185	578.1	Included only nut spread food codes.
Candy Containing Chocolate	185	616.7	Included all candy containing chocolate food codes.

g – Grams; mg - Milligrams

Table 2. 2017-2020 NHANES Food Codes Used for Intake Analysis

Food Code	Main Food Description
Sports and Energy Drinks	
95310200	Energy drink (Full Throttle)
95310400	Energy drink (Monster)
95310500	Energy drink (Mountain Dew AMP)
95310550	Energy drink (No Fear)
95310555	Energy drink (No Fear Motherload)
95310560	Energy drink (NOS)
95310600	Energy drink (Red Bull)
95310700	Energy drink (Rockstar)
95310750	Energy drink (SoBe Energize Energy Juice Drink)
95310800	Energy drink (Vault)
95311000	Energy Drink
95312560	Energy drink (Ocean Spray Cran-Energy Juice Drink)
95312900	Energy drink (XS)
95312905	Energy drink (XS Gold Plus)
95320200	Sports drink (Gatorade G)
95320500	Sports drink (Powerade)
95321000	Sports drink, NFS
Diet Sports and Energy Drinks	
95312400	Energy drink, low calorie (Monster)
95312410	Energy drink, sugar free (Monster)
95312500	Energy drink, sugar free (Mountain Dew AMP)
95312550	Energy drink, sugar free (No Fear)
95312555	Energy drink, sugar-free (NOS)
95312600	Energy drink, sugar-free (Red Bull)
95312700	Energy drink, sugar free (Rockstar)
95312800	Energy drink, sugar free (Vault)

Food Code	Main Food Description
95313200	Energy drink, sugar free
95322200	Sports drink, low calorie (Gatorade G2)
95322500	Sports drink, low calorie (Powerade Zero)
95323000	Sports drink, low calorie
Ready-to-eat Cereal (Low Sugar)	
57000100	Cereal, oat, NFS
57101000	Cereal (Kellogg's All-Bran)
57103000	Cereal (Post Alpha-Bits)
57106050	Cereal (Post Great Grains Banana Nut Crunch)
57123000	Cereal, O's, plain
57132000	Cereal, corn squares
57134000	Cereal, corn flakes, plain
57135000	Cereal (Kellogg's Corn Flakes)
57137000	Cereal, corn puffs
57148000	Cereal (Kellogg's Crispix)
57151000	Cereal, rice crispy, plain
57206700	Cereal (General Mills Fiber One)
57206710	Cereal (General Mills Fiber One Honey Clusters)
57207000	Cereal, bran flakes, plain
57208000	Cereal (Kellogg's All-Bran Complete Wheat Flakes)
57209000	Cereal (Post Bran Flakes)
57214000	Cereal, shredded wheat, flavored
57228000	Granola, homemade
57230000	Cereal (Post Grape-Nuts)
57237100	Cereal, oat bunches
57237200	Cereal (Post Honey Bunches of Oats with Vanilla Bunches)
57237300	Cereal (Post Honey Bunches of Oats with Almonds)
57241200	Cereal (Post Shredded Wheat Honey Nut)
57301500	Cereal (Kashi 7 Whole Grain Puffs)
57301505	Cereal (Kashi Autumn Wheat)
57301510	Cereal (Kashi GOLEAN)
57301530	Cereal (Kashi Heart to Heart Honey Toasted Oat)
57303100	Cereal (General Mills Kix)
57303105	Cereal (General Mills Honey Kix)
57304100	Cereal, oat squares
57305160	Cereal (Malt-O-Meal Blueberry Muffin Tops)
57306700	Cereal (Malt-O-Meal Toasted Oat Cereal)
57308400	Cereal, O's, multigrain
57321900	Cereal (Nature's Path Organic Flax Plus)
57326000	Cereal (Barbara's Puffins)
57327450	Cereal (Quaker Toasted Oat Bran)
57327500	Cereal (Quaker Oatmeal Squares)

Food Code	Main Food Description
57336000	Cereal, rice squares
57337000	Cereal, rice flakes
57339000	Cereal (Kellogg's Rice Krispies)
57340000	Cereal, puffed rice
57344000	Cereal, K's, plain
57401100	Cereal, O's, NFS
57408100	Cereal (Uncle Sam)
57411000	Cereal, wheat squares
57416000	Cereal, plain puffs
57417000	Cereal, shredded wheat, plain
57418000	Cereal, wheat flakes
57301600	Cereal, multigrain
57420110	Cereal, other, plain
57420160	Cereal, reduced sugar
Yogurt (Regular)	
11400000	Yogurt, NFS
11410000	Yogurt, NS as to type of milk or flavor
11411010	Yogurt, NS as to type of milk, plain
11411100	Yogurt, whole milk, plain
11411200	Yogurt, low fat milk, plain
11411300	Yogurt, nonfat milk, plain
11430000	Yogurt, NS as to type of milk, fruit
11431000	Yogurt, whole milk, fruit
11433000	Yogurt, nonfat milk, fruit
11434090	Yogurt, NS as to type of milk, flavors other than fruit
11434100	Yogurt, whole milk, flavors other than fruit
11434200	Yogurt, low fat milk, flavors other than fruit
11434300	Yogurt, nonfat milk, flavors other than fruit
11436000	Yogurt, liquid
11446000	Yogurt parfait, low fat, with fruit
42401100	Yogurt, coconut milk
Yogurt (Greek)	
11400010	Yogurt, Greek, NS as to type of milk or flavor
11411390	Yogurt, Greek, NS as to type of milk, plain
11411400	Yogurt, Greek, whole milk, plain
11411410	Yogurt, Greek, low fat milk, plain
11411420	Yogurt, Greek, nonfat milk, plain
11433990	Yogurt, Greek, NS as to type of milk, fruit
11434000	Yogurt, Greek, whole milk, fruit
11434010	Yogurt, Greek, low fat milk, fruit
11434020	Yogurt, Greek, nonfat milk, fruit
11435000	Yogurt, Greek, NS as to type of milk, flavors other than fruit

Food Code	Main Food Description
11435010	Yogurt, Greek, whole milk, flavors other than fruit
11435020	Yogurt, Greek, low fat milk, flavors other than fruit
11435030	Yogurt, Greek, nonfat milk, flavors other than fruit
11435100	Yogurt, Greek, with oats
Milk (Lowfat)	
11115400	Kefir, NS as to fat content
Cereal Bars	
53710400	Cereal or granola bar (General Mills Fiber One Chewy Bar)
53710500	Cereal or granola bar (Kellogg's Nutri-Grain Cereal Bar)
53710502	Cereal or granola bar (Kellogg's Nutri-Grain Yogurt Bar)
53710504	Cereal or granola bar (Kellogg's Nutri-Grain Fruit and Nut Bar)
53710600	Milk 'n Cereal bar
53710700	Cereal or granola bar (Kellogg's Special K bar)
53710810	Cereal or granola bar (KIND Fruit and Nut Bar)
53710900	Cereal or granola bar (General Mills Nature Valley Chewy Trail Mix)
53710902	Cereal or granola bar, with yogurt coating (General Mills Nature Valley Chewy Granola Bar)
53710904	Cereal or granola bar (General Mills Nature Valley Sweet and Salty Granola Bar)
53710906	Cereal or granola bar (General Mills Nature Valley Crunchy Granola Bar)
53711000	Cereal or granola bar (Quaker Chewy Granola Bar)
53711002	Cereal or granola bar (Quaker Chewy 90 Calorie Granola Bar)
53711004	Cereal or granola bar (Quaker Chewy 25% Less Sugar Granola Bar)
53711006	Cereal or granola bar (Quaker Chewy Dipp's Granola Bar)
53711100	Cereal or granola bar (Quaker Granola Bites)
53712000	Snack bar, oatmeal
53712100	Cereal or Granola bar, NFS
53712200	Cereal or granola bar, lowfat, NFS
53712210	Cereal or granola bar, nonfat
53713000	Cereal or granola bar, reduced sugar, NFS
53713010	Cereal or granola bar, fruit and nut
53713100	Cereal or granola bar, peanuts , oats, sugar, wheat germ
53714200	Cereal or granola bar, chocolate coated, NFS
53714210	Cereal or granola bar, with coconut, chocolate coated
53714220	Cereal or granola bar with nuts, chocolate coated
53714230	Cereal or granola bar, oats, nuts, coated with non-chocolate coating
53714250	Cereal or granola bar, coated with non-chocolate coating
53714300	Cereal or granola bar, high fiber, coated with non-chocolate yogurt coating
53714400	Cereal or granola bar, with rice cereal
53714500	Breakfast bar, NFS
53714510	Breakfast bar, date, with yogurt coating
53714520	Breakfast bar, cereal crust with fruit filling, lowfat
Nutrition Bars	

Food Code	Main Food Description
53710800	Cereal or granola bar (Kashi Chewy)
53710802	Cereal or granola bar (Kashi Crunchy)
53720100	Nutrition bar (Balance Original Bar)
53720200	Nutrition bar (Clif Bar)
53720210	Nutrition bar (Clif Kids Organic Zbar)
53720300	Nutrition bar (PowerBar)
53720400	Nutrition bar (Slim Fast Original Meal Bar)
53720500	Nutrition bar (Snickers Marathon Protein Bar)
53720600	Nutrition bar (South Beach Living Meal Bar)
53720610	Nutrition bar (South Beach Living High Protein Bar)
53720700	Nutrition bar (Tiger's Milk)
53720800	Nutrition bar (Zone Perfect Classic Crunch)
53729000	Nutrition bar or meal replacement bar, NFS
Ice Cream and Frozen Desserts	
11459990	Frozen yogurt, NFS
11460000	Frozen yogurt, vanilla
11460100	Frozen yogurt, chocolate
11460500	Frozen yogurt, soft serve, vanilla
11460510	Frozen yogurt, soft serve, chocolate
11461200	Frozen yogurt sandwich
11461210	Frozen yogurt bar, vanilla
11461220	Frozen yogurt bar, chocolate
11461250	Frozen yogurt cone, chocolate
11461260	Frozen yogurt cone, vanilla
11461300	Frozen yogurt cone, vanilla, waffle cone
11461320	Frozen yogurt cone, chocolate, waffle cone
13110000	Ice cream, NFS
13110100	Ice cream, vanilla
13110102	Ice cream, vanilla, with additional ingredients
13110110	Ice cream, chocolate
13110112	Ice cream, chocolate, with additional ingredients
13110200	Ice cream, soft serve, vanilla
13110210	Ice cream, soft serve, chocolate
13110460	Gelato, vanilla
13110470	Gelato, chocolate
13120050	Ice cream bar, vanilla
13120100	Ice cream bar, vanilla, chocolate coated
13120110	Ice cream candy bar
13120140	Ice cream bar, chocolate
13120500	Ice cream sandwich, vanilla
13120510	Ice cream sandwich, chocolate
13120550	Ice cream cookie sandwich

Food Code	Main Food Description
13120730	Ice cream cone, scooped, vanilla
13120735	Ice cream cone, scooped, vanilla, waffle cone
13120740	Ice cream cone, NFS
13120770	Ice cream cone, scooped, chocolate
13120775	Ice cream cone, scooped, chocolate, waffle cone
13120782	Ice cream cone, soft serve, vanilla
13120784	Ice cream cone, soft serve, chocolate
13120786	Ice cream cone, soft serve, vanilla, waffle cone
13120788	Ice cream cone, soft serve, chocolate, waffle cone
13120790	Ice cream cone, vanilla, prepackaged
13120792	Ice cream cone, chocolate, prepackaged
13121000	Ice cream sundae, NFS
13121100	Ice cream sundae, fruit topping
13121120	Banana split
13121300	Ice cream sundae, hot fudge topping
13121400	Ice cream sundae, caramel topping
13126000	Ice cream, fried
13130100	Light ice cream, NFS
13130300	Light ice cream, vanilla
13130310	Light ice cream, chocolate
13130700	Soft serve, blended with candy or cookies, from fast food
13135000	Light ice cream sandwich, vanilla
13135010	Light ice cream sandwich, chocolate
13140000	Light ice cream bar, vanilla
13140100	Light ice cream bar, vanilla, chocolate coated
13140115	Light ice cream bar, chocolate
13140700	Creamside
13140710	Creamside, light
13140900	Fudgesicle
13142100	Light ice cream cone, vanilla, prepackaged
13142110	Light ice cream cone, chocolate, prepackaged
13150000	Sherbet, all flavors
13161600	Fudgesicle, light
Nuts and Seeds	
42202000	Peanut butter
42202010	Peanut butter, lower sodium
42202100	Peanut butter, lower sodium and lower sugar
42202130	Peanut butter, lower sugar
42202150	Peanut butter, reduced fat
42202200	Peanut butter, vitamin and mineral fortified
42203000	Peanut butter and jelly
42203100	Peanut butter and chocolate spread

Food Code	Main Food Description
Candy Containing Chocolate	
91700500	M&M's Almond Chocolate Candies
91701010	Almonds, chocolate covered candy
91703040	Caramel candy, chocolate covered
91703050	Caramel with nuts and cereal, chocolate covered
91703060	Caramel with nuts, chocolate covered
91703070	Rolo
91703150	Toblerone, milk chocolate with honey and almond nougat
91703200	TWIX Caramel Cookie Bars
91703250	TWIX Chocolate Fudge Cookie Bars
91703300	TWIX Peanut Butter Cookie Bars
91703400	Whatchamacallit
91703600	Espresso coffee beans, chocolate-covered
91705010	Chocolate candy
91705020	Chocolate candy with cereal
91705030	Kit Kat
91705040	Chocolate, milk, with nuts, not almond or peanuts
91705050	Milk chocolate candy, with fruit and nuts
91705060	Milk chocolate candy, with almonds
91705070	Chocolate, milk, with peanuts
91705090	Chocolate candy with fondant and caramel
91705200	Chocolate chips
91705300	Dark chocolate candy
91705310	Chocolate, sweet or dark, with almonds
91705500	Mexican chocolate, tablet
91706000	Coconut candy, chocolate covered
91707010	Fondant, chocolate covered
91709000	Gumdrops, chocolate covered
91713010	Fudge, chocolate, chocolate-coated
91713020	Fudge, chocolate, chocolate-coated, with nuts
91713030	Fudge, chocolate
91713040	Fudge, chocolate, with nuts
91715000	Fudge, caramel and nut, chocolate-coated candy
91715100	SNICKERS Bar
91715200	Baby Ruth
91715300	100 GRAND Bar
91716110	Halvah, chocolate covered
91718050	Honey-combed hard candy with peanut butter, chocolate covered
91718100	Butterfinger
91718110	Butterfinger Crisp
91718200	Chocolate-flavored sprinkles
91723010	Marshmallow, chocolate covered

Food Code	Main Food Description
91726110	Nougat, with caramel, chocolate covered
91726130	MILKY WAY Bar
91726140	MILKY WAY MIDNIGHT Bar
91726150	MARS Almond Bar
91726410	Nougat, chocolate covered
91726420	3 MUSKETEERS Bar
91726425	3 Musketeers Truffle Crisp Bar
91727010	Nuts, chocolate covered, not almonds or peanuts
91731000	Peanuts, chocolate covered candy
91731010	M&M's Peanut Chocolate Candies
91731060	M&M's Peanut Butter Chocolate Candies
91733200	Peanut Bar, chocolate covered candy
91734000	Peanut butter, chocolate covered
91734100	Reese's Peanut Butter Cup
91734300	Reese's Sticks
91734400	Reese's Fast Break
91734450	Reese's Crispy Crunchy Bar
91739010	Raisins, chocolate covered
91746010	Sugar-coated chocolate discs
91746100	Chocolate candy, candy shell
91746120	Sixlets
91746150	Easter egg, candy coated chocolate
91746200	M&M's Pretzel Chocolate Candies
91760100	Toffee, chocolate covered
91760200	Toffee, chocolate-coated, with nuts
91760500	Truffles
91770030	Dietetic or low calorie candy, chocolate covered
91705005	Chocolate candy, other, NFS
91705012	Chocolate candy with nuts, other, NFS
91705015	Chocolate candy with nuts
91705080	Chocolate candy, cookie filled
91705290	Dark chocolate candy, other, NFS
91705312	Dark chocolate candy with nuts, other, NFS
91705315	Dark chocolate candy with nuts
91705440	Chocolate candy, fudge
91705450	Chocolate candy, caramel filled
91705460	Chocolate candy, caramel filled with nuts
91705470	Chocolate candy, coconut filled
91705480	Chocolate candy, cream filled
91705510	Chocolate candy, nougat filled
91705520	Chocolate candy, nougat filled with nuts
91705530	Chocolate candy, peanut butter filled

Food Code	Main Food Description
91705550	Chocolate candy with dried fruit
91706010	Chocolate candy, sugar free
91746110	Chocolate candy, candy shell with nuts

Estimated Daily Intake (EDI) of *P. goldsteinii* RV-01 from Proposed Uses

Based on the proposed uses, the following estimated daily intake data was calculated in absolute, milligrams per day, (Table 3) and relative to consumers' body weight (Table 4) for both Total Population (Per Capita) and Consumers Only.

Table 3. Estimated Daily Intake of *P. goldsteinii* RV-01 (mg/day)

Population Group	Number of Users	Percentage (%)	Ingredient Intake – Per Capita (mg/day)				Ingredient Intake – Consumer-Only (mg/day)			
			Mean	±SE	90th Percentile	±SE	Mean	±SE	90th Percentile	±SE
1-5 y, F	331	66%	99.4	6.4	252.1	12.8	147.0	6.6	277.5	20.1
1-5 y, M	339	62%	107.9	7.8	258.9	17.9	158.9	8.3	342.7	30.5
6-11 y, F	408	68%	135.7	7.1	335.2	25.6	186.3	6.8	375.3	19.2
6-11 y, M	366	61%	113.8	7.1	315.2	31.9	176.8	7.6	403.6	32.7
12-18 y, F	372	57%	154.8	11.1	394.2	42.6	246.5	9.8	581.1	46.7
12-18 y, M	360	54%	133.9	11.3	406.1	39.3	238.7	12.5	513.9	43.4
19+ y, F	2011	57%	119.7	3.6	322.1	9.3	190.7	3.5	402.5	11.8
19+ y, M	1743	54%	169.4	6.2	471.8	17.0	279.1	7.1	618.5	23.4
1+ y, M+F	5930	57%	139.7	2.7	375.0	6.9	223.8	2.5	475.9	5.4

SE – Standard error; y – Years; M – Males; F – Females; mg – Milligrams

Table 4. Estimated Daily Intake of *P. goldsteinii* RV-01 (mg/kg bw/day)

Population Group	Number of Users	Percentage (%)	Ingredient Intake – Per Capita (mg/kg bw/day)				Ingredient Intake – Consumer-Only (mg/kg bw/day)			
			Mean	±SE	90th Percentile	±SE	Mean	±SE	90th Percentile	±SE
1-5 y, F	331	66%	6.3	0.4	15.3	0.9	9.3	0.5	18.2	1.3
1-5 y, M	339	62%	6.7	0.5	17.5	1.3	9.8	0.6	19.8	1.7
6-11 y, F	408	68%	4.1	0.3	10.5	0.6	5.6	0.2	11.9	0.6

6-11 y, M	366	61%	3.6	0.3	9.6	1.1	5.6	0.3	13.1	0.9
12-18 y, F	372	57%	2.7	0.2	6.6	0.8	4.2	0.2	10.4	0.7
12-18 y, M	360	54%	2.1	0.2	6.4	0.5	3.7	0.2	8.2	0.5
19+ y, F	2011	57%	1.7	0.1	4.6	0.2	2.7	0.1	5.9	0.1
19+ y, F	1743	54%	1.9	0.1	5.3	0.3	3.2	0.1	7.1	0.3
1+ y, M+F	5930	57%	2.3	0.1	6.3	0.1	3.7	0.1	8.2	0.2

SE – Standard error; y – Years; M – Males; F – Females; mg – Milligrams; kg – Kilograms; bw – Body weight;

¹ A note on Creme Food Safety® model: The system supports both deterministic (single points) input data and probabilistic input data. Probabilistic data can be represented by parametric or empirical distributions, integrated in the analysis using Monte Carlo simulations. If the type of modelling used is probabilistic, it means the input data may be represented as probability distributions when required (rather than using point estimates only) and the simulation process uses conditional distribution sampling. When using probabilistic modelling and a population-based approach, the output from the model – the systemic exposure results – are themselves a distribution. In the Creme Food Data Science ® model, standard errors of statistics are calculated using a resampling technique called bootstrapping. For example, a mean value can be estimated from the collected sample data which is assumed to be representative of the total population. Using the bootstrap method allows a distribution of the mean values to be generated and used to assess the accuracy of the estimated statistic (in this case, the mean value). This is performed by sampling with replacement from the data set in question several times, generating a number of different estimates of each statistic. The standard error of the mean is then the standard deviation of the mean values obtained from the large number of bootstrap samples.

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Appendix 7 GRAS Associates Expert Panel Report

The Generally Recognized as Safe (GRAS) Status of the Proposed Uses of Autoclaved *Parabacteroides goldsteinii* RV-01

April 22, 2025

Foreword

An independent panel of experts (“Expert Panel”) was convened by GRAS Associates, LLC on behalf of their client, ReviveBio, Co., to evaluate the safety and Generally Recognized as Safe (GRAS) status of ReviveBio, Co.’s proposed uses of Autoclaved *Parabacteroides goldsteinii* RV-01 in conventional foods. The members of this Expert Panel[†] are qualified to serve in this capacity by qualification of scientific training and experience in the safety of food and food ingredients.

GRAS Associates and ReviveBio, Co. ensured that all reasonable efforts were made to identify and select a balanced Expert Panel with expertise in food safety, toxicology, and nutrition. The Expert Panel was selected and convened in accordance with the Food and Drug Administration (FDA)’s guidance for industry on “Best Practices for Convening a GRAS Panel”²⁹. Efforts were placed on identifying conflicts of interest or relevant “appearance issues” that could potentially bias the outcome of the deliberations of the Expert Panel and no such conflicts of interest or “appearance issues” were identified. The Expert Panel members received a reasonable honorarium as compensation for their time; the honoraria provided to the Expert Panel members were not contingent upon the outcome of their deliberations.

Discussion

Autoclaved *P. goldsteinii* RV-01 is the result of heat-inactivation of *P. goldsteinii* RV-01, a strictly anaerobic, gram-negative bacterium that is endogenously found in the human gastrointestinal tract and is non-pathogenic and non-toxigenic. The strain was identified via 16S rRNA sequencing, whole genome sequencing, and average nucleotide identity analysis.

The manufacturing process of Autoclaved *P. goldsteinii* RV-01 is performed according to current Good Manufacturing Practices (cGMP) and utilizes food grade ingredients. The process includes the seeding and growth of live bacterium in fermentation tanks, its harvest and centrifugation, followed by autoclaving, mixing with maltodextrin, lyophilization, milling and storage. Genetic integrity of the strain

[†] Dr. Preece, Chair of the Expert Panel, is a naturopathic physician and board-certified toxicologist with food safety experience. Dr. Gregor Reid is Distinguished Professor Emeritus at Western University in Canada and international expert on ‘biotics’. Dr. Daniel O’Sullivan is a Professor in the Food Science & Nutrition Department at the University of Minnesota with over 30 experience researching probiotic and gut microorganisms.

²⁹ Available at: <https://www.fda.gov/Food/GuidanceRegulation/GuidanceDocumentsRegulatoryInformation/ucm583856.htm>

is ensured over repeated manufacturing processes through the use of a 16S rRNA sequencing-validated working cell bank. Batch consistency is ensured with specifications that include qualitative and quantitative identification parameters of the bacterium in addition to contaminants such as heavy metal and other microorganisms. The efficacy of autoclaving is ensured with a parameter limiting the number of viable *P. goldsteinii* RV-01 to “not detected”. Five non-consecutive batches were provided as evidence of production consistency.

Autoclaved *P. goldsteinii* RV-01 is intended to be used in sports and energy drinks, ready-to-eat cereals (low sugar), yogurt, fermented milk products, ice cream, cereal bars, nutrition bars, nut spreads, and candy containing chocolate at levels up to 1×10^{10} total fluorescent units (TFU)/serving. It is not intended to be used in infant formula or USDA regulated foods and will not be marketed to children. Under the intended conditions of use, the estimated mean and 90th percentile intakes in the general population 1.2×10^{10} TFU/day (2.0×10^8 TFU/kg bw/day) and 2.6×10^{10} TFU/day (4.4×10^8 TFU/kg bw/day), respectively.

Safety information on Autoclaved *P. goldsteinii* RV-01 is generally available. Although genotypic profiling of live *P. goldsteinii* RV-01 showed evidence of the tetQ gene, which is associated with tetracycline resistance, the lack of its DNA in Autoclaved *P. goldsteinii* RV-01, showed by PCR, suggested DNA is degraded during the manufacturing process. The strain does not contain genomic elements for secondary metabolites of concern or biogenic amines. Toxicological studies, including GLP- and OECD-compliant genotoxicity studies (bacterial reverse mutation assay, in vitro chromosomal aberration, and in vivo micronucleus assay) and subchronic, repeated-dose, oral toxicity studies (28- and 90-day rodent studies) have shown Autoclaved *P. goldsteinii* RV-01 has no toxicological concern under the conditions tested. Specifically, the no observed adverse effect level in the 90-day, repeated-dose toxicological study was 1,500 mg/kg bw/day (8.1×10^{10} TFU/kg bw/day). Based on this NOAEL, the margin of safety is 183 in the general population. Therefore, the composite evidence from historical safe consumption and experimental studies collectively demonstrates the safety of Autoclaved *P. goldsteinii* RV-01 for human food consumption at the above-described intended use levels.

In summary, sufficient qualitative and quantitative scientific evidence in the composite is available to support the safety-in-use of ReviveBio, Co.’s Autoclaved *P. goldsteinii* RV-01 preparation given the following conditions:

ReviveBio, Co.’s Autoclaved *P. goldsteinii* RV-01 preparation continues to meet the designated specifications; and

ReviveBio, Co.’s Autoclaved *P. goldsteinii* RV-01 preparation is produced in accordance with Current Good Manufacturing Practices (CGMPs).

Conclusion

The Expert Panel critically reviewed the data provided by ReviveBio, Co. for their Autoclaved *P. goldsteinii* RV-01 preparation, as well as publicly available published information obtained from peer-reviewed journals and other safety assessments prepared by other Expert Panels and well-respected international regulatory bodies.

The ingestion of ReviveBio, Co.'s Autoclaved *P. goldsteinii* RV-01 from the intended uses results in intakes that are safe within the limits of established historical use and published safety studies.

The Expert Panel unanimously concluded that the proposed uses of ReviveBio, Co.'s Autoclaved *P. goldsteinii* RV-01 preparation, manufactured as described in Part 2 of their dossier, and declared within the subject notification meets the FDA definition of safety in that there is "reasonable certainty of no harm under the intended conditions of use" as described herein, and ReviveBio, Co.'s Autoclaved *P. goldsteinii* RV-01 preparation is generally recognized as safe (GRAS).

It is also our opinion that other qualified and competent scientists reviewing the same publicly available toxicological and safety information would reach the same conclusion. Therefore, we have also concluded that Autoclaved *P. goldsteinii* RV-01, when used as described in this dossier, is GRAS based on scientific procedures.



Kayla Preece, ND, DABT
Chair



Daniel J. O'Sullivan, PhD



Gregor Reid, PhD, MBA, ARM CCM, Dr HS,
FCAHS, FRSC

END

DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved:OMB No. 0910-0342; Expiration Date: 08/31/2025 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER	DATE OF RECEIPT
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____ ☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3 Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): _____	
4 For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person Hsin-Chih Lai		Position or Title General Manager	
	Organization (<i>if applicable</i>) ReviveBio, Co.			
	Mailing Address (<i>number and street</i>) 18F-1, No. 3, Park St., NangGang District			
City Taipei City		State or Province Taiwan		Zip Code/Postal Code 11503
Country China				
Telephone Number 02-26557778		Fax Number 02-26557779		E-Mail Address hsin.lai@revivebio.com.tw
1b. Agent or Attorney (<i>if applicable</i>)	Name of Contact Person William J. Rowe		Position or Title President & CEO	
	Organization (<i>if applicable</i>) GRAS Associates			
	Mailing Address (<i>number and street</i>) 11810 Grand Park Avenue, Suite 500			
City North Bethesda		State or Province Maryland		Zip Code/Postal Code 20852
Country United States of America				
Telephone Number 772-532-3454		Fax Number		E-Mail Address amozingo@gras-associates.com

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Autoclaved Parabacteroides goldsteinii RV-01

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway ☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)* ☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN _____

☐ b) GRAS Affirmation Petition No. GRP _____

☐ c) Food Additive Petition No. FAP _____

☐ d) Food Master File No. FMF _____

☐ e) Other or Additional *(describe or enter information as above)* _____

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))* ☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? (see 21 CFR 170.225(c)(8))

☐ Yes *(Proceed to Item 8)*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

ReviveBio, Co. autoclaved P. goldsteinii RV-01 as defined in the subject notification is expected to be safe for use as an ingredient in sports and energy drinks, ready-to-eat cereals (low sugar), yogurt, fermented milk products, ice cream, cereal bars, nutrition bars, nut spreads, and candy containing chocolate at levels up to 1×10^{10} TFU/serving, or 185 mg/serving. The intended use does not include infant formula or USDA-regulated food. The mean and 90th percentile intakes of autoclaved P. goldsteinii RV-01 in the general population are 223.8 mg/day (1.0×10^{10} TFU/day) and 475.9 mg/day (2.2×10^{10} TFU/day), respectively.

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☐ Yes ☒ No

Did you include this other information in the list of attachments?

☐ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that ReviveBio, Co.

(name of notifier)

has concluded that the intended use(s) of Autoclaved Parabacteroides goldsteinii RV-01

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. ReviveBio, Co. agrees to make the data and information that are the basis for the
(name of notifier) conclusion of GRAS status available to FDA if FDA asks to see them;
agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

18F-1, No. 3, Park St., NangGang District Taipei City, Taiwan 11503 China

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Amy Mozingo

Digitally signed by Amy Mozingo
Date: 2025.05.06 11:05:07 -04'00'

Printed Name and Title

Amy Mozingo on behalf of William J. Rowe, Agent

Date (mm/dd/yyyy)

05/06/2025

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	ReviveBio_GRAS_P.goldsteiniiRV-01_22April2025(1).pdf	Administrative
	G10922_ReviveBio_P.goldsteiniiRV-01_TransmittalLetter.pdf	Administrative

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRASStaff@fda.hhs.gov. (Please do NOT return the form to this address.). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.